Cross spectra between thin redshift bins

Which pairs the data vector measures, and how the
Fingers-of-God blur changes their overlap

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1 Summary

The questions. The forecast cuts the DESI galaxies into thin redshift bins and measures the angular power spectrum of every bin. For some pairs of bins it also measures their cross spectrum, the correlation between the two bins' maps. Four bin layouts slice the galaxies: $\Delta d_{\rm com} = 62$ Mpc, 76 bins of equal comoving depth, and three of fixed redshift width, $\Delta z = 0.02$, 0.05 and 0.2, with 90, 36 and 10 bins. The first two are the thin layouts. This note answers three questions:

1. How does the Fingers-of-God (FoG) blur, at its default strength, change how much neighbouring bins overlap?
2. Which pairs of bins have their cross spectrum in the data vector, and why those?
3. What does that rule mean in redshift, on each of the four bin layouts?

The answers.

1. The blur has a width of 5–6 Mpc along the line of sight, about a tenth of a thin bin. It raises the overlap of two neighbouring thin bins from about 0.003, the one grid step where their tabulated edges meet, to about 0.06. Bins one apart stay disjoint. On $\Delta d_{\rm com} = 62$ Mpc the blur makes ten LRG3–ELG1 pairs overlap by 0.03, half as much as a neighbour pair, although their raw ranges just miss; the data vector measures them too (Figure 1).
2. A pair is measured when the two bins' redshift distributions $n(z)$ overlap or touch, either as measured or once blurred by the FoG effect of item 1. That gives the neighbours along each galaxy sample's chain of bins, plus the pairs of an LRG bin and an ELG bin at about the same redshift: at $0.8 < z < 1.1$ the LRG3 and ELG1 groups are two galaxy samples observing one volume (Figures 2 and 3). The four layouts measure 107, 133, 52 and 13 cross spectra.
3. On the layouts of fixed redshift width only adjacent bins pair. On $\Delta d_{\rm com} = 62$ Mpc the rule is adjacency in comoving distance, which no single cut in redshift can express: a bin of fixed comoving depth is 0.015 wide in redshift at $z = 0.1$ and 0.035 at $z = 1.5$.

Beyond the current list. A further forecast measures what the pairs outside this rule would add, along two axes: pairs whose raw ranges are up to three bin widths apart, and pair lists selected with every dispersion raised to a common test value (Section 12). Those pairs have no kernel overlap; their signal is the low-ℓ correlation that survives only outside the

Limber approximation (Section 9). On the three thinner layouts those pairs raise the paper model's dark-energy figure of merit FoM(w_0, w_a) – the inverse area of the error ellipse of the dark-energy parameters w_0 and w_a, defined in Section 11 – by up to 55–59%, on $\Delta z = 0.2$ by at most 2%. Measured against the wide layout itself, the thin slicing is worth far more than the added pairs: at $k_{\max} = 0.2\,h/\mathrm{Mpc}$ the two thin layouts' FoM(w_0, w_a) is 22–26 times that of $\Delta z = 0.2$ on the current lists, and the added pairs raise the factors to 28–37 (Section 12). Attribution fits trace the gain to the free level and slope of each sample group's bias amplitude $\beta_i = b_{1,i}\, D(\bar{z}_i)$, the linear bias times the growth factor D, the amplitude of matter clustering at the bin's epoch relative to today: in the auto spectra the level is degenerate with the clustering amplitude and the slope with the evolution of growth that w_0 and w_a control – σ_v rides on the same degeneracy – while the cross spectra's correlation coefficients do not depend on the linear bias, so the added pairs separate the bias line from the growth history; with every nuisance parameter held fixed they are worth only 2–8% (Section 13). Every bias prior that passes the project's accuracy test gives the same gain within a few percent; under a prior on the bias levels and slopes themselves it would be smaller, by an amount not yet computed. The analysis keeps the current list; the longer lists are a comparison against that standard choice. The gain and a magnification sensitivity travel together. The model sets the magnification bias b_{mag}, the response of the observed number density to the lensing convergence, to zero, and the added pairs' neglected magnification signal looks to the fit like a change of those same bias levels and slopes: at the measured count slopes the neglect shifts the fits by at most about 0.07 of the marginalized errors on the current and the longer lists alike, while at an unfavourable value of the unmeasured ELG count slope the largest lists reach 0.30–0.46 where the current lists stay below 0.06 (Section 12; `notes/lensing_magnification`).

2 Fiducial model and parameter values

Three of this note's figures (Figures 1, 2 and 3) depend only on the shipped redshift distributions, the pair lists and the default velocity dispersions: no scale cut and no bias value enters them. The toy of Figure 4 uses stand-in numbers of its own, stated in Section 9; the measured curves of Figure 5 are the fiducial spectra of the pair-axes forecasts and one Limber evaluation of the same model. Only the Fisher forecasts of Sections 11 and 12 run a fit. A Fisher forecast predicts the error bars a data set will deliver from the derivatives of the model spectra and from the covariance, without running chains; `notes/fisher_machinery` builds that method from scratch. This section freezes the values behind both fits, so the record does not depend on configuration files that can change.

Geometry of the pair rule and the masks. The comoving distances of the $\Delta d_{\mathrm{com}} = 62$ Mpc slicing, of the pair rule and of the shipped masks – the files that list which data-vector entries a fit keeps – use $\chi(z)$ of a flat ΛCDM with $H_0 = 67.32$ km/s/Mpc and $\Omega_m = 0.3$; the masks stop every bin at $\ell + 1/2 \le k\,\chi(\bar{z}_i)$ with $k = 0.1185\,\mathrm{Mpc}^{-1}$ (about $0.176\,h/\mathrm{Mpc}$), with no growth scaling.

Layouts and pairs. The four layouts hold 76, 90, 36 and 10 lens bins and measure 107, 133, 52 and 13 cross pairs ($\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2; Tables 1 and 3). The paper data sets carry these pair lists; the exploration examples keep the older 97-pair `*_linear` files on $\Delta d_{\mathrm{com}} = 62$ Mpc (Section 6).

Velocity dispersions. The fiducial dispersions are 371, 358, 357, 348, 343, 304 and 289 km/s for the seven groups (BGS split at $\bar{z} = 0.255$), with exploration prior widths 145, 105, 70, 50, 50, 40 and 30 km/s and paper widths six times these (Table 2 and Section 5). Every blur in this note runs at the fiducials, except the pair selection on the σ_v axis of Section 12; the fit of Section 11 uses the exploration widths, the forecasts of Section 12 the paper widths.

The pair-threshold fit. Its cosmology is the evaluate override of the example yamls, the values the example configuration files fix when they evaluate one model instead of sampling: $10^9 A_s = 2.1$ (A_s the amplitude of the primordial curvature power spectrum), $n_s = 0.96605$, $H_0 = 67.32$ km/s/Mpc, $\Omega_b = 0.04$, $\Omega_m = 0.3$, $w_0 = -1$, $w_a = 0$, $\sum m_\nu = 0.06$ eV and $\tau = 0.0697186$ fixed; the derived $\sigma_8 = 0.8278$. The linear bias is $b_{1,i} = B_g / D(\bar{z}_i)$ with $B_g = 1.34$ (BGS), 1.70 (LRG1–LRG3) and 0.84 (ELG1, ELG2), under the pessimistic prior (3% correlated residuals, $L = 0.05$, a free amplitude and tilt per group; `notes/correlated_bias_prior_v2`). The higher-order terms are those of the fiducial model, the one-loop terms beyond the linear bias that `notes/higher_order_bias` defines one by one: the offsets Δ_2 and Δ_s of the quadratic and tidal biases from their calibrations, of prior widths 2 and 1; the counterterms α_0 and α_2, k^2 terms that absorb small-scale physics the perturbative model cannot predict (α_2 on the redshift-space part), of width $0.5/k_{\rm max}^2$ at each cut; the stochastic amplitudes SN_0 and SN_2 of widths 2 and 5; and flat tilts, with α_2 free and the exploration σ_v widths: the reference model of `notes/fog_parameter_degeneracy_study`, not the paper model, which fixes $\alpha_2 = 0$ in every bin and widens every σ_v prior sixfold. The pair-axes forecasts of Section 12 run the same cosmology and scale cut with the paper model.

Scale cut and covariance of the fit. The fit keeps the bands of the non-Limber UV-fraction cut of Saraivanov et al. [9], computed on the paper model's fiducial at the final derivative tier (`chains/model_scale_cuts_final.npz`; a band is kept while less than 5% of it comes from the power spectrum above $k_{\rm max}/D(\bar{z}_i)$), intersected with the shipped mask. The covariance is the Gaussian covariance of the 107-pair data set – the covariance the band powers would have if the maps were Gaussian random fields, computed from the model spectra and the shot noise – built on the one-loop fiducial, the fiducial point of the one-loop bias model (b_2 and b_{s^2} on their calibrations), with massless neutrinos over 14,000 deg^2. The forecast sky is 12,300 deg^2, and a Gaussian covariance scales inversely with the area (more sky, more independent modes), so the inverse covariance is rescaled by 12,300/14,000. The forecasts of Section 12 build the same Gaussian covariance on each layout's largest pair list.

3 Why cross spectra between thin bins matter

The angular power spectrum. Take the galaxies of bin i and draw a map of their number on the sky: $\delta_i(\hat{n})$ is the fractional excess of galaxies in the direction $\hat{n}$ over the mean. Expand the map in spherical harmonics, $\delta_i(\hat{n}) = \sum_{\ell m} a^i_{\ell m} Y_{\ell m}(\hat{n})$. The multipole ℓ labels an angular scale of about $180^\circ/\ell$: $\ell = 20$ is about 9°, $\ell = 500$ about 0.4°.

The auto spectrum of bin i is the variance of its coefficients, $C^{ii}_\ell = \langle |a^i_{\ell m}|^2 \rangle$; the cross spectrum of bins i and j is their covariance, $C^{ij}_\ell = \langle a^i_{\ell m} a^{j*}_{\ell m} \rangle$. Two unrelated maps have a cross spectrum that averages to zero; two maps of the same structures have a positive one. The data vector holds these spectra in 53 bands of width $\Delta\ell = 10$ over $15 \le \ell \le 544$: a band averages

the spectrum over ten consecutive multipoles, and one band of one spectrum is one entry of the data vector.

Why neighbouring thin bins correlate. A thin bin is a shell of the universe a few tens of Mpc deep: 62 Mpc on $\Delta d_{\rm com} = 62$ Mpc, and between 37 and 85 Mpc for $\Delta z = 0.02$. The matter distribution has structures of every size. One that is 200 Mpc across spans three or four such shells, and each of them sees it in the same direction on the sky. Their maps share it, so the cross spectrum of two neighbouring shells is large at the angular scale of that structure. A structure a few Mpc across sits in one shell only, unless it sits where two shells meet.

Limber and non-Limber. These two cases are the two regimes of a cross spectrum. At high ℓ the Limber approximation [1] holds: only Fourier modes perpendicular to the line of sight contribute, and

$$C_\ell^{ij} \;\simeq\; \int d\chi\, \frac{W_i(\chi)\, W_j(\chi)}{\chi^2}\, P\!\left(k = \frac{\ell + 1/2}{\chi}\right), \tag{1}$$

with χ the comoving distance, $W_i(\chi)$ the redshift distribution of bin i written per unit comoving distance – the bin's *kernel*, the name used from here on – and $P(k)$ the power spectrum of the galaxy density at wavenumber k. Two shells whose W do not overlap then have no cross spectrum. At low ℓ the approximation fails for thin shells: modes with a component along the line of sight longer than the shell depth connect shells that do not touch. The forecast computes every spectrum without the Limber approximation, so it keeps that low-ℓ correlation. Section 9 works the difference out step by step on bins with zero overlap.

Where the radial information sits. Slicing a sample into thin shells keeps information along the line of sight: a shell of depth $\Delta\chi$ resolves line-of-sight wavenumbers up to about $\pi/\Delta\chi$, $0.05\,\mathrm{Mpc}^{-1}$ for 62 Mpc. The auto spectra say how clustered each shell is; the cross spectra say how the clustering of one shell continues into the next.

How much the cross spectra are worth depends on the bin layout: the thinner the shells, the more of the line-of-sight information sits in the cross spectra between neighbours rather than in the auto spectra.

4 The six groups and the LRG–ELG overlap

Three galaxy samples. DESI selects three kinds of galaxies, each by its own colours and brightness, and each fills the redshifts where it is bright and numerous enough to observe:

- **BGS**, the Bright Galaxy Survey: the brightest galaxies at low redshift, here $0.1 < z < 0.4$;
- **LRG**, luminous red galaxies: massive galaxies that stopped forming stars, $0.4 < z < 1.1$;
- **ELG**, emission-line galaxies: star-forming galaxies, found by the bright emission lines of their gas, $0.8 < z < 1.6$.

Nothing forces these ranges to tile the redshift axis. LRG and ELG both cover $0.8 < z < 1.1$.

Six groups. The forecast splits the samples into six groups by the bins' mean redshift $\bar{z}_i$, and gives each group its own bias parameters (Table 1): BGS whole; LRG1 ($\bar{z}_i < 0.6$), LRG2 ($0.6 \le \bar{z}_i < 0.8$) and LRG3 ($\bar{z}_i \ge 0.8$); ELG1 ($\bar{z}_i < 1.1$) and ELG2 ($\bar{z}_i \ge 1.1$). LRG3 and ELG1 are therefore two different galaxy populations in the same volume, at $z \approx 0.8$–1.1.

group	sample	bins	$\bar{z}_i$
BGS	bright galaxies	1–18	0.11–0.38
LRG1	luminous red galaxies	19–29	0.41–0.60
LRG2	luminous red galaxies	30–38	0.62–0.79
LRG3	luminous red galaxies	39–49	0.81–1.06
ELG1	emission-line galaxies	50–61	0.81–1.09
ELG2	emission-line galaxies	62–76	1.11–1.55

Table 1: The six groups of $\Delta d_{\rm com} = 62$ Mpc, with bins numbered from 1 in data-vector order. LRG3 and ELG1 cover the same redshifts.

The numbering runs group by group. Bins are numbered in the order of the data vector: all BGS bins by increasing redshift, then all LRG bins, then all ELG bins. The numbering therefore jumps back in redshift between bins 49 and 50: bin 49 (LRG3) has $\bar{z} = 1.060$, bin 50 (ELG1) has $\bar{z} = 0.812$. Two bins far apart in number can sit at the same redshift: LRG3 bin 44 has $\bar{z} = 0.932$ and ELG1 bin 55 has $\bar{z} = 0.931$.

What the LRG–ELG cross spectrum measures. An LRG bin and an ELG bin at the same redshift sample the same matter with different galaxies. On large scales a galaxy map is the matter map times a number, the linear bias b_i of the bin, so their cross spectrum is about $b_i b_j$ times the matter spectrum, while the two autos are about b_i^2 and b_j^2 times it. The three spectra together tie the two biases to each other. Two tracers of the same structures also share the random fluctuations of those structures, so comparing them measures their bias ratio with less of that randomness than either map alone (the multi-tracer idea [4]).

Shot noise and the stochastic terms. A galaxy map is made of discrete objects, so even a random scatter of galaxies has a nonzero auto spectrum: the shot noise, $1/\bar{n}_i$ for a bin of $\bar{n}_i$ galaxies per steradian. A cross spectrum has no shot noise, because no galaxy is in two bins: within one sample each galaxy has one measured redshift, and two samples are different galaxies.

The one-loop model – the perturbation-theory model of the galaxy spectrum one order beyond the linear bias of the previous paragraph, whose terms `notes/higher_order_bias` defines – adds stochastic terms beyond this Poisson noise, SN_0 and SN_2. They are properties of the three-dimensional galaxy field, so they reach the cross spectrum of two bins of one sample through the overlap of their kernels, $\int d\chi\, W_i W_j/\chi^2$ for SN_0 and the same integral of the kernels' radial derivatives for SN_2 (`stochastic_templates` in `likelihood/desi_cl.py`). An LRG–ELG pair gets none: two different populations do not share a stochastic term, and the model has no separate cross-stochasticity parameter.

5 The Fingers-of-God blur

5.1 What it is

Redshift is not distance. The forecast places each galaxy by its measured redshift. That redshift has two parts: the cosmological redshift, set by the galaxy's distance, and a Doppler shift from the galaxy's own velocity v along the line of sight, its peculiar velocity. The two multiply, $1 + z_{\rm obs} = (1 + z)(1 + v/c)$, so for $v \ll c$

$$z_{\rm obs} - z \;\simeq\; (1 + z)\,\frac{v}{c}. \tag{2}$$

For example, $v = 350$ km/s at $z = 0.5$ shifts the redshift by 1.75×10^{-3}. In distance this is $\delta\chi = c\,\delta z/H(z) = (1 + z)\,v/H(z) \approx 6$ Mpc, with $H(0.5) = 88$ km/s/Mpc in the examples' cosmology ($H_0 = 67.32$ km/s/Mpc, $\Omega_m = 0.3$).

Fingers of God. Inside groups and clusters, galaxies orbit their common halo with random speeds of several hundred km/s. In a map that places every galaxy at its redshift distance, a cluster is stretched along the line of sight into a long finger pointing at the observer: the Fingers of God [5]. The orderly infall of galaxies onto large structures is a different effect, the Kaiser distortion [6], which the model computes separately.

What it does to a thin bin. A bin collects the galaxies whose measured redshift falls inside its range: a sharp cut in $z_{\rm obs}$. The matter that the galaxies trace sits at their true distances. A galaxy just outside the range, moving towards it, is measured inside; one just inside, moving away, is measured outside. So the galaxies of the bin trace the matter over a slightly wider, softer-edged range than the histogram of their measured redshifts. The same convolution describes a sharp shell in true redshift observed blurred along the line of sight; for a symmetric velocity distribution the two descriptions agree.

5.2 The model in the likelihood

The likelihood replaces each bin's raw $n_i(z)$ by its convolution with a Gaussian in redshift [8],

$$n_i^{\rm FoG}(z) = \int dz'\, n_i(z')\,\frac{1}{\sqrt{2\pi}\,\sigma_{z,i}}\,\exp\left[-\frac{(z - z')^2}{2\sigma_{z,i}^2}\right], \qquad \sigma_{z,i} = (1 + \bar{z}_i)\,\frac{\sigma_v}{c}, \tag{3}$$

with σ_v the velocity dispersion of the bin's group, the standard deviation of the line-of-sight velocities, and $c = 299792.458$ km/s. The width follows from Eq. (2).

The code is `set_lens_related` in `likelihood/desi_cl.py`. It convolves a copy of each raw histogram with `scipy.ndimage.gaussian_filter1d` on the $n(z)$ table's grid, the DESI grid refined five-fold (step 7.0×10^{-5}, so $\sigma_{z,i}$ spans 20 to 36 steps), with the width in grid steps and zero beyond the grid ends. The Gaussian is truncated at $4\sigma_{z,i}$, scipy's default. The σ_v are sampled parameters, so the convolution runs again at every sampled point. The figures of this note repeat that call; at the default σ_v the result equals, bit for bit, the convolved tables that the covariance adapter writes with the same call (`covariance/<layout>_fog.nz`).

5.3 Where the velocity dispersions come from

The source forecast measured the line-of-sight velocity dispersion of its simulated galaxies as a function of redshift, with an error bar per bin. The project reads one value per group off that measurement (Table 2). The measured curve steps inside BGS, between the bins at $\bar{z} = 0.247$ and 0.263, so BGS takes two values, split at $z = 0.255$: seven dispersion groups for six bias groups.

group	σ_v [km/s]	exploration width	paper width	$\sigma_{z,i}$ [10^{-3}]	$\delta\chi$ [Mpc]
BGS, $\bar{z}_i < 0.255$	371	145	870	1.37–1.54	5.8–6.1
BGS, $\bar{z}_i \geq 0.255$	358	105	630	1.51–1.65	5.9–6.0
LRG1	357	70	420	1.68–1.90	6.0–6.1
LRG2	348	50	300	1.88–2.08	6.0
LRG3	343	50	300	2.07–2.36	5.8–5.9
ELG1	304	40	240	1.84–2.12	5.1–5.2
ELG2	289	30	180	2.04–2.46	4.6–4.8

Table 2: The default velocity dispersion of each group, the widths of its Gaussian prior in the exploration files and in the paper model (six times the exploration width; both in km/s), and the resulting redshift width $\sigma_{z,i}$ and comoving width $\delta\chi = (1 + \bar{z}_i)\,\sigma_v/H(\bar{z}_i)$ over the bins of $\Delta d_{\rm com} = 62$ Mpc.

A prior is the knowledge the fit assumes about a parameter before seeing the data; a Gaussian prior of width 145 km/s penalizes a sampled dispersion once it departs from the default by about that much. Each prior here is a Gaussian centred on the default and truncated at zero, since a negative dispersion has no meaning. The exploration width is five times the per-bin error bar of the measurement; the paper model multiplies every width by six, so that the sampled dispersions can absorb the radial counterterm it drops (`notes/fog_parameter_degeneracy_study`). At the sixfold width the truncation matters for the first BGS group: it removes 33% of the Gaussian, and the truncated mean is about 850 km/s, while the Fisher forecasts treat every σ_v prior as an untruncated Gaussian about the default. The exploration defaults and widths live in `likelihood/params_desi*.yaml` (the `DESI_SIGV` parameters; the first bin of each group carries the prior and the others are tied to it), the paper widths in `likelihood/params_desi*_bias_pessimistic.yaml`, and the same seven defaults in `CHUNK_SIGMA_V` of `covariance/lssty1xdesi_covariance.py`, which convolves the covariance's kernels at fixed σ_v. Every figure of this note blurs at the defaults, except that the σ_v axis of Section 12 selects pairs with the blur at a common test dispersion; the fit of Section 11 uses the exploration widths (Section 2). The ELG dispersions are the lowest, as expected for star-forming galaxies, which tend to live in lighter halos with slower orbits.

5.4 How large the blur is

A nearly constant width in distance. In comoving distance the blur is $\delta\chi = (1 + z)\,\sigma_v/H(z)$ (last column of Table 2): 5.8–6.1 Mpc for BGS and LRG and 4.6–5.2 Mpc for ELG, at every redshift. It barely changes with z because $(1 + z)/H(z)$ changes by at most 10% over $0.1 < z < 1.6$.

Against the bin width. What matters for overlap is $\sigma_{z,i}$ divided by the bin's width w_i, the redshift range its galaxies were selected in. On $\Delta d_{\rm com} = 62$ Mpc it is the same ratio in distance, $\delta\chi/62$ Mpc: 0.069–0.096, about a tenth of the bin at every redshift. On $\Delta z = 0.02$ it grows with redshift, from 0.069 to 0.125, because a fixed Δz is a thinner shell at high z. On $\Delta z = 0.05$ it is 0.028–0.050, and on $\Delta z = 0.2$ 0.007–0.023 (Figure 1, top right).

A Gaussian blur moves galaxies across a sharp edge: a galaxy a distance x inside the edge leaves with the probability that its blur exceeds x, and integrating that Gaussian tail over the bin moves a fraction of about $\sigma_z/(\sqrt{2\pi}\,w)$ of the bin's galaxies across each edge. The blurred $n(z)$ of a thin bin has 5–8% of its weight outside its raw range on $\Delta d_{\rm com} = 62$ Mpc and 5–10% on $\Delta z = 0.02$, against 2–4% on $\Delta z = 0.05$ and 0.5–2% on $\Delta z = 0.2$. That weight lies inside the neighbouring bins.

5.5 What it does to the overlap

A measure of overlap. For two bins define the normalized overlap integral

$$O_{ij} \;=\; \frac{\int dz\, n_i(z)\, n_j(z)}{\sqrt{\int dz\, n_i^2(z)\, \int dz\, n_j^2(z)}}, \tag{4}$$

1 for two identical distributions and 0 for disjoint ones, whatever their normalization. It has a direct meaning. In the Limber regime, Eq. (1), the power spectrum and χ barely change across two thin bins, so O_{ij} is close to the correlation coefficient $C_\ell^{ij}/\sqrt{C_\ell^{ii}C_\ell^{jj}}$ of the two bins at high ℓ. The SN_0 cross term of Section 4 is nearly the same integral.

Neighbours. Without the blur, two neighbouring bins touch only over the one grid step ($h = 3.5\times10^{-4}$ in redshift) where their tabulated edges meet. The tables are read with linear interpolation, so across that step one bin falls from its height to zero while the other rises; the product of the falling and the rising ramp integrates to $h/6$ ($\int_0^1 t\,(1-t)\,dt = 1/6$, in units of h and of the heights), while the normalization of Eq. (4) contributes about the bin width w. The overlap is therefore about $h/(6\,w)$: 0.0015–0.0039 on $\Delta d_{\rm com} = 62$ Mpc, 0.0027–0.0029 on $\Delta z = 0.02$, 0.0011–0.0012 on $\Delta z = 0.05$ and 0.0003–0.0004 on $\Delta z = 0.2$. This number measures the grid, not the galaxies. With the blur at the default σ_v the edges become smooth ramps about $4\sigma_z$ wide. Where the bins meet, the falling ramp of one times the rising ramp of the other now integrates to $\sigma_z/\sqrt{\pi} \approx 0.56\,\sigma_z$, so the overlap becomes about $0.6\,\sigma_z/w$: 0.039–0.061 on $\Delta d_{\rm com} = 62$ Mpc (median 0.057), 0.043–0.083 on $\Delta z = 0.02$ (0.066), 0.018–0.032 on $\Delta z = 0.05$ (0.024) and 0.006–0.009 on $\Delta z = 0.2$ (0.007). On the thin layouts this is a 6% correlation at high ℓ that the raw $n(z)$ would put at 0.3%.

Bins one apart stay disjoint. The bin between them is at least $8\sigma_z$ wide, and the Gaussian is truncated at $4\sigma_z$, so the blurred $n(z)$ of two bins one apart do not meet. The only exception is the last ELG2 bins of $\Delta z = 0.02$, where $\sigma_z/w = 0.125$ and the truncated tails touch, with an overlap below 2×10^{-14}. Even an untruncated Gaussian gives two blurred top hats one bin apart a percent-level overlap only at $\sigma_z \approx 0.4\,w$, four times the default on the thin layouts. Within two prior widths of every default σ_z/w stays below 0.2, where that overlap is below 2×10^{-5}.

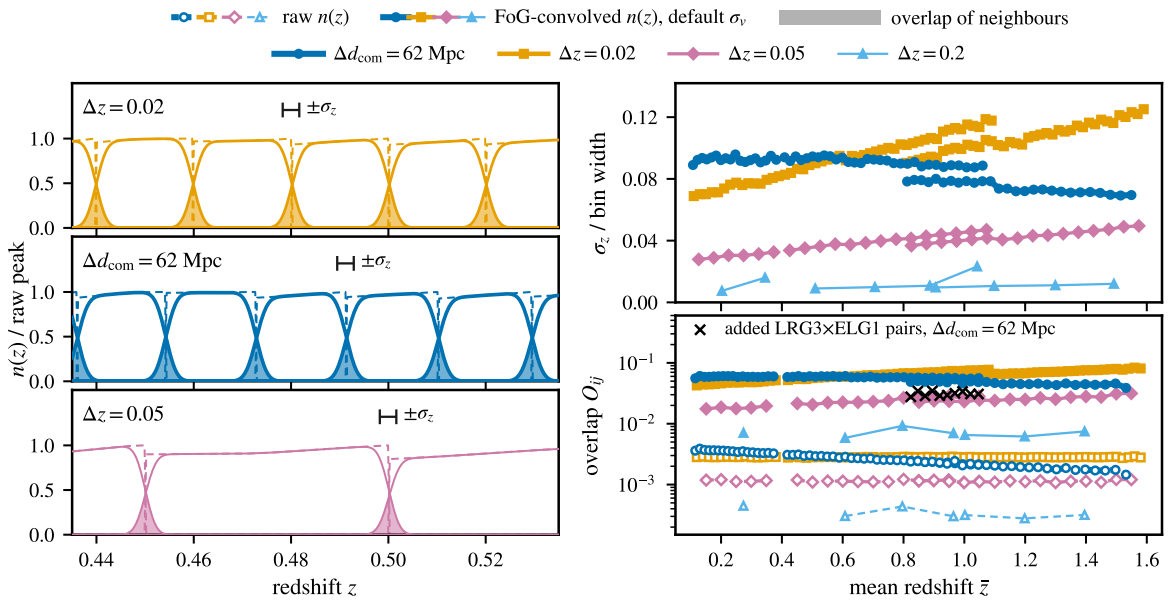


Figure 1: The Fingers-of-God blur at the default σ_v. Left: raw (dashed) and blurred (solid) $n(z)$ of the LRG1 bins in $0.435 < z < 0.535$ on three layouts, each divided by its raw peak; shaded: the overlap $\min(n_i, n_j)$ of neighbouring blurred bins; bar: $\pm\sigma_z$. Right: σ_z over the bin width, and the overlap O_{ij} of Eq. (4) for consecutive bins of one sample, raw (dashed, open markers) and blurred (solid, filled), against the mean redshift. Crosses: the ten LRG3–ELG1 pairs of $\Delta d_{\rm com} = 62$ Mpc that the second step of the pair rule (Section 6) adds, whose raw overlap is zero. Bins one apart overlap by at most 2×10^{-14}, raw or blurred.

6 Which pairs are measured

The first step: raw ranges. Each bin's raw $n(z)$ is the histogram of the measured redshifts of its galaxies, tabulated on the DESI redshift grid of step $h = 3.5 \times 10^{-4}$. Let $z_i^{\rm lo}$ and $z_i^{\rm hi}$ be the first and last grid redshifts where $n_i(z)$ is nonzero. Bins i and j pair when their ranges overlap or touch:

$$z_j^{\rm lo} \le z_i^{\rm hi} + 1.5\,h \quad \text{and} \quad z_i^{\rm lo} \le z_j^{\rm hi} + 1.5\,h. \tag{5}$$

The slack of 1.5 grid steps is there because two neighbouring bins of one sample either share their edge node or sit one node apart; 1.5 steps catches both and nothing farther.

For example, on $\Delta z = 0.02$ bin 1 ends at $z = 0.1198$ and bin 2 starts one grid step later, at 0.1202: they pair. Bin 3 starts at 0.1401, 0.020 after the end of bin 1: bins 1 and 3 do not pair.

The second step: blurred kernels. The model never uses the raw $n(z)$ directly. It first blurs each one by the Fingers-of-God effect of Section 5, which spreads every bin a few Mpc beyond its edges. Two bins whose raw ranges just miss can therefore still overlap once blurred. The second step adds every such pair: bins i and j also pair when their blurred $n(z)$ overlap at all, $O_{ij} \ge 10^{-6}$ at the default velocity dispersions, with O_{ij} the normalized overlap integral of Eq. (4) (`scripts/make_desi_pairs.py`).

The threshold is not a tuning. Before this step the blurred overlaps of unmeasured pairs fall into two sets far apart: ten pairs at 0.028–0.035, and every other pair below 2×10^{-14}, the reach of the truncated Gaussian tails. Any threshold between 10^{-6} and 0.0277 gives the same lists: those ten pairs on $\Delta d_{\rm com} = 62$ Mpc, none on the fixed-Δz layouts.

Where the pair lists come from. For the three fixed-Δz layouts the first step writes the pair list (`cross_pairs` in `scripts/make_desi_dz_case.py`), and the second adds nothing. On $\Delta d_{\rm com} = 62$ Mpc the first list comes from the DESI forecast's data file (a `sacc` file, a standard container format for survey data vectors), which pairs bins by redshift proximity; `scripts/make_desi_dataset.py` imports it. The first step reproduces 96 of its 97 pairs. The exception is the boundary pair of BGS bin 18 and LRG1 bin 19: BGS ends at $z = 0.390$ and LRG starts at $z = 0.400$, so their ranges are 0.010 apart, and the forecast includes them anyway; the list keeps it. Blurred, their kernels overlap by only 3×10^{-7}: the ranges are six times σ_z apart. The second step adds ten LRG3–ELG1 pairs (Section 7), for 107 pairs in all.

The $\Delta d_{\rm com} = 62$ Mpc data vector. With the added pairs the $\Delta d_{\rm com} = 62$ Mpc data vector holds 198 spectra: 15 cosmic-shear spectra, 76 autos and 107 cross spectra, in 53 bands each, 10,494 entries of which the mask keeps 5,674. The cosmic-shear spectra are the correlations of the measured shapes of the LSST source galaxies, which gravitational lensing distorts; the data vector carries them alongside the DESI clustering spectra ($15 \times 53 = 795$ entries), but every forecast of this note masks them and uses the clustering spectra alone, and the pair rule concerns only the clustering part. Each added pair keeps the bands where both of its autos are kept, the rule every other pair follows. The Gaussian covariance rebuilt on this layout is positive definite and bitwise equal to the 97-pair covariance on the shared spectra. The exploration examples keep the 97-pair files (the `*_linear` dataset, mask, pair list and covariance), so their χ^2 values stay as committed.

The counts. Table 3 sorts every layout's pairs by kind. Most join a bin to the next one of its sample. The rest join an LRG and an ELG bin: the LRG3–ELG1 pairs of the shared volume (on $\Delta d_{\rm com} = 62$ Mpc, 22 from the first step and 10 from the second), one LRG2–ELG1 pair at $z = 0.8$ where LRG2 ends and ELG1 begins, and, on $\Delta z = 0.02$ and 0.05, one LRG3–ELG2 pair at $z = 1.1$. Each bin has 1 to 5 cross partners, 1 to 4 on $\Delta z = 0.2$; the first and last bins have one.

layout	bins	next bin	BGS–LRG1	LRG3–ELG1	LRG–ELG, other	added	total
$\Delta d_{\rm com} = 62$ Mpc	76	73	1	22	1	10	107
$\Delta z = 0.02$	90	87	1	43	2	0	133
$\Delta z = 0.05$	36	33	1	16	2	0	52
$\Delta z = 0.2$	10	7	1	4	1	0	13

Table 3: The measured cross spectra of each layout, by kind. "Next bin": two consecutive bins of one sample; BGS–LRG1: the pair across the BGS–LRG boundary at $z = 0.4$; "LRG–ELG, other": the LRG2–ELG1 pair at $z = 0.8$ and the LRG3–ELG2 pair at $z = 1.1$; "added": the LRG3–ELG1 pairs of the second step, whose raw ranges miss and whose blurred $n(z)$ overlap.

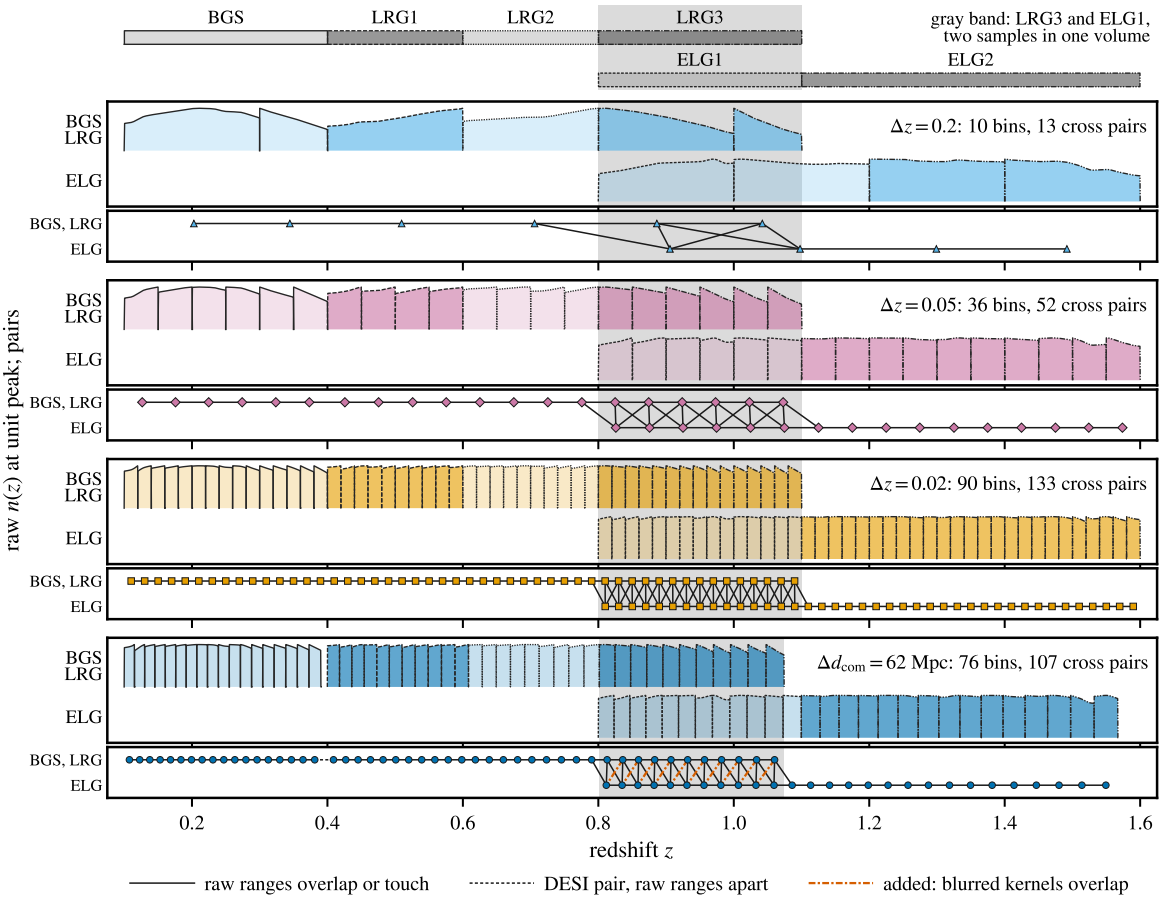


Figure 2: Every thin bin of the four layouts and the cross spectra the data vector measures. Upper panel of each block: each bin's raw $n(z)$ at unit peak, BGS and LRG in the upper lane, ELG in the lower; fill shade and outline style mark the six groups (header). Lower panel: a dot per bin at its mean redshift and a segment per measured pair; solid: raw ranges overlap or touch; dashed: the BGS–LRG1 pair of $\Delta d_{\rm com} = 62$ Mpc, whose ranges do not touch; dash-dotted vermillion: the ten LRG3–ELG1 pairs added because their blurred $n(z)$ overlap. Gray band: the redshifts where LRG3 and ELG1 both have bins.

How to read Figure 2. Each block is one layout. Its upper panel draws every bin's raw $n(z)$ at unit peak, BGS and LRG in the upper lane, ELG in the lower one; two bins that touch share an edge. Its lower panel is the pair graph: a dot per bin at $\bar{z}_i$ and a segment per measured pair. The segments inside a row are the chain of neighbours; the segments between the rows are the LRG–ELG pairs, all inside the gray band where LRG3 and ELG1 share a volume, or at its edges. The dash-dotted vermillion segments of $\Delta d_{\rm com} = 62$ Mpc are the ten pairs of the second step. On $\Delta z = 0.02$ the LRG and ELG bins have the same edges, so each LRG3 bin touches three ELG1 bins: the one at its redshift and the two beside it. Figure 3 magnifies the corner where LRG2 ends and LRG3 and ELG1 begin.

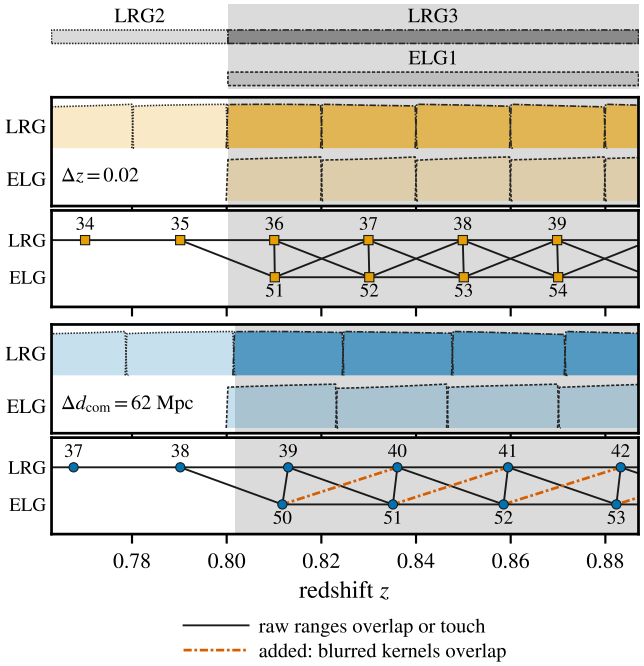


Figure 3: The corner of Figure 2 where LRG2 ends and LRG3 and ELG1 begin, on the two thin layouts, with every bin numbered in data-vector order. Solid: measured pairs whose raw ranges overlap or touch. Dash-dotted vermillion: the LRG3–ELG1 pairs of $\Delta d_{\rm com} = 62$ Mpc whose raw ranges miss by 3 to 5 grid steps and which the second step adds because their blurred $n(z)$ overlap (Section 5). A segment leaving the panel joins a bin outside it.

7 What the rule means in redshift

Centre separations. A rule in terms of ranges can be compared with a rule in terms of centres: is there a separation $|\bar{z}_i - \bar{z}_j|$ below which every pair is measured and above which none is? Table 4 gives, per layout, the largest separation of a measured pair and the smallest of an unmeasured one.

layout	bin width in z	largest measured	smallest unmeasured
$\Delta z = 0.2$	0.1–0.2	0.211 (LRG3 5, ELG1 8)	0.257 (LRG3 6, ELG2 9)
$\Delta z = 0.05$	0.05	0.053 (BGS 6, LRG1 7)	0.097 (LRG3 18, ELG1 22)
$\Delta z = 0.02$	0.02	0.020 (LRG3 48, ELG1 64)	0.040 (LRG3 45, ELG1 58)
$\Delta d_{\rm com} = 62$ Mpc	0.015–0.035	0.035 (ELG2 75, 76)	0.030 (BGS 1, 3)

Table 4: The separation $|\bar{z}_i - \bar{z}_j|$ of the mean redshifts of the farthest measured pair and of the closest unmeasured pair, with their groups and bin numbers. On $\Delta z = 0.2$ the bins at the ends of some samples are 0.1 wide.

Fixed redshift width: only adjacent bins. On the three fixed-Δz layouts the two columns are separated by about a factor of two: neighbours are one width apart, the next bin over is two widths apart. Any cut between the two columns, for example at 0.03 on $\Delta z = 0.02$,

reproduces the rule. So on these layouts the rule means: only adjacent bins pair, where an LRG bin and an ELG bin with the same range count as adjacent (separation near zero). The edge bins of $\Delta z = 0.2$, 0.1 wide, shift the numbers but not the conclusion.

Fixed comoving width: no redshift cut works. On $\Delta d_{\rm com} = 62$ Mpc the columns cross: a measured pair is 0.035 apart and an unmeasured one 0.030. The reason is how a shell of fixed depth looks in redshift. The comoving distance is $\chi(z) = \int_0^z c\,dz'/H(z')$, with $H(z)$ the expansion rate, so a shell of depth $\Delta d_{\rm com}$ spans $\Delta z \approx \Delta d_{\rm com}\,H(z)/c$ in redshift. H grows with z, so the same 62 Mpc covers 0.015 in redshift at $\bar{z} = 0.11$ and 0.035 at $\bar{z} = 1.55$.

Two examples make this concrete. BGS bins 1 and 3, one bin apart at $z \approx 0.12$, are only 0.030 apart in mean redshift and not paired; ELG2 bins 75 and 76, neighbours at $z \approx 1.53$, are 0.035 apart and paired. The rule is adjacency in comoving distance, and in redshift that adjacency stretches with z.

Misaligned LRG and ELG edges. The LRG3–ELG1 pairs of $\Delta d_{\rm com} = 62$ Mpc show a second effect. Each sample has its own bin edges: the LRG bins begin at $z = 0.400$ and the ELG bins at $z = 0.800$. In the shared volume the LRG edges sit 0.001–0.002 above the ELG edges (Figure 3, bottom). An LRG3 bin therefore overlaps the ELG1 bin at its redshift and the one above it, and starts just above the end of the one below it. Bin 40 begins 0.0014 above the end of bin 50: 4 grid steps, more than the first step's 1.5, so the raw ranges miss, although the centres are only 0.024 apart. The blur closes that gap: the ten pairs of an LRG3 bin and the ELG1 bin ending just below it (bins 40 to 49 with 50 to 59) have raw ranges 3 to 5 grid steps apart, less than σ_z, and zero raw overlap, but blurred they overlap by 0.028–0.035, half as much as a neighbour pair (crosses in Figure 1; dash-dotted segments in Figures 2 and 3). At high ℓ they therefore correlate about half as strongly as neighbours do, and the second step of the rule adds them to the data vector. Each of the LRG3 bins 40 to 49 thus has three ELG1 partners, as on $\Delta z = 0.02$: two from the first step and one from the second, 32 LRG3–ELG1 pairs in all. With the ten added, no unmeasured pair of any layout overlaps by more than 10^{-6} after the blur; on the fixed-Δz layouts none did to begin with.

8 Which $n(z)$ each piece reads

The pair list reads the raw $n(z)$ in its first step, Eq. (5), and the blurred $n(z)$ at the default σ_v in its second. It is fixed with the data and does not move when σ_v is sampled. The model reads the blurred $n(z)$ in every spectrum it computes: the autos, the measured cross spectra and the stochastic templates. The covariance uses the blurred $n(z)$ at the default σ_v.

Together the two steps measure every pair whose kernels overlap in the model, and the DESI forecast's BGS–LRG1 boundary pair besides. On the fixed-Δz layouts the second step changes nothing: every pair whose blurred $n(z)$ overlap already touches in raw $n(z)$, and every measured pair overlaps by at least 0.003 once blurred.

9 Why bins with zero overlap still correlate

The question. The pair rule of Section 6 keeps a cross spectrum when the two bins' $n(z)$ overlap, raw or blurred. That invites a question: if two bins share no redshift at all, not

even blurred, how can their maps be correlated? In the Limber approximation, Eq. (1), they cannot: the integrand holds the product $W_i W_j$, which is zero at every χ. But the forecast computes its spectra without that approximation (Section 3), and the exact cross spectrum of two separated bins is small but not zero. This section builds the difference step by step, first on a toy with explicit numbers (Figure 4), then on a measured pair of the $\Delta z = 0.02$ layout (Figure 5).

Separated shells still hold galaxy pairs. The configuration-space version of the cross spectrum makes the physics plain. Let $\xi(r)$ be the correlation function of the galaxy density: the average product of the density contrast at two points a three-dimensional distance r apart, positive where structures of that size are common. The angular correlation of the maps of bins i and j, at separation θ on the sky, sums ξ over every pair of one point in each shell:

$$w_{ij}(\theta) = \int d\chi \int d\chi'\, n_i(\chi)\, n_j(\chi')\, \xi(r), \qquad r = \sqrt{\chi^2 + \chi'^2 - 2\chi\chi' \cos\theta}, \tag{6}$$

with $n_i(\chi)$ the bin's kernel per unit comoving distance. Two shells whose centres are 110 Mpc apart hold pairs at $r = 110$ Mpc and beyond, and ξ is not zero there. Nothing in the exact expression asks the shells to overlap. The overlap only takes over when ξ is treated as so short-ranged that only the pairs with $\chi = \chi'$ count.

The exact projection. In harmonic space the same statement reads

$$C_\ell^{ij} = \frac{2}{\pi} \int dk\, k^2\, P(k)\, W_i^\ell(k)\, W_j^\ell(k), \qquad W_i^\ell(k) = \int d\chi\, n_i(\chi)\, b_i\, D(\chi)\, j_\ell(k\chi), \tag{7}$$

with $P(k)$ the matter power spectrum today, b_i the linear bias of bin i and $D(\chi)$ the linear growth factor ($D = 1$ today), which carry the matter clustering to the galaxies at the shell's epoch, and j_ℓ the spherical Bessel function, the radial oscillation that a plane wave of wavenumber k shows when expanded in spherical harmonics around the observer. The kernel $W_i^\ell(k)$ says how strongly a Fourier mode of wavenumber k imprints on the map of bin i. Two separated shells respond to the same long mode, so the product $W_i^\ell W_j^\ell$ need not vanish anywhere.

The toy. Figure 4 works this out for the simplest case: two top-hat shells of width $w = 55$ Mpc at comoving distance $\chi = 3200$ Mpc, the size and distance of one $\Delta z = 0.02$ ELG bin at $z = 0.93$, with the smooth BBKS power spectrum [3] of shape $\Gamma = \Omega_m h$ ($\Omega_m = 0.315$, $h = 0.674$, $n_s = 1$, no baryon features). The toy is flat-sky, so it is inexact below $\ell \approx 10$, and its sharp edges and smooth spectrum are idealizations; the measured pair below needs none of them.

The flat-sky form. On a flat sky each Fourier mode splits into a transverse wavenumber $k_\perp$, seen on the sky at multipole $\ell = k_\perp \chi$, and a line-of-sight wavenumber $k_\parallel$. Up to a factor common to every spectrum,

$$C^{ij}(\ell) \propto \int_{-\infty}^{\infty} \frac{dk_\parallel}{2\pi}\, P_s(k_\perp, k_\parallel)\, \mathrm{Re}\big[W_i(k_\parallel)\, W_j^*(k_\parallel)\big], \qquad k_\perp = \frac{\ell}{\chi}, \tag{8}$$

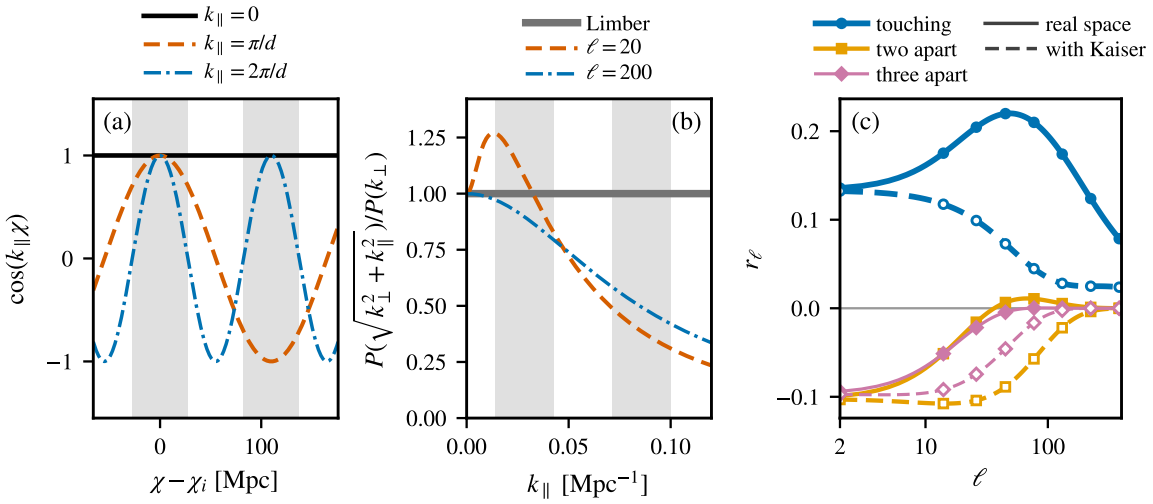


Figure 4: The toy of Section 9: two top-hat shells of width 55 Mpc at $\chi = 3200$ Mpc with the BBKS power spectrum. (a) The shells two bins apart (gray bands, centres $d = 110$ Mpc apart) and three line-of-sight modes $\cos(k_\parallel \chi)$: $k_\parallel = 0$ and $2\pi/d$ take the same value in both shells, π/d opposite values. (b) The power a mode carries over the one power the Limber approximation gives every mode, at $\ell = 20$ and 200; shaded: the $k_\parallel$ where the weight of Eq. (9) is negative; gray: the Limber value, one. (c) The correlation coefficient r_ℓ of touching shells and of shells two and three bins apart, in real space (solid, filled markers) and with the linear Kaiser distortion (dashed, open); color and marker: the separation; gray: the Limber value, zero for all three, since no pair of shells overlaps.

where $W_i(k_\parallel) = \int d\chi\, n_i(\chi)\, e^{ik_\parallel \chi}$ is the Fourier transform of the shell profile along the line of sight and $P_s(k_\perp, k_\parallel)$ is the power of the mode: in real space $P(k)$ with $k = \sqrt{k_\perp^2 + k_\parallel^2}$, in redshift space the direction-dependent form below. For two top-hat shells of width w whose centres are a distance d apart the weight is

$$\mathrm{Re}\big[W_i W_j^*\big] = \cos(k_\parallel d)\,\mathrm{sinc}^2\left(\frac{k_\parallel w}{2}\right), \qquad \mathrm{sinc}\, x = \frac{\sin x}{x}. \tag{9}$$

The line-of-sight modes, one by one. Equation (8) is a sum over line-of-sight modes, each with the weight of Eq. (9). Panel (a) of Figure 4 draws three modes across two shells two bins apart, $d = 110$ Mpc. The mode $k_\parallel = 0$ is constant along the line of sight: it takes the same value in both shells, weight $+1$. So does $k_\parallel = 2\pi/d$, one full wavelength from centre to centre. The mode $k_\parallel = \pi/d$ puts a crest in one shell and a trough in the other, weight -1: where that mode makes one shell overdense it makes the other underdense. Without overlap the positive and negative weights balance exactly,

$$\int_{-\infty}^{\infty} \frac{dk_\parallel}{2\pi}\, \mathrm{Re}\big[W_i W_j^*\big] = \int d\chi\, n_i(\chi)\, n_j(\chi) = 0 \tag{10}$$

(Parseval's theorem; the right-hand side is the unnormalized overlap of the two kernels).

What Limber assumes, and why it gives zero. The Limber approximation [1, 2] treats the power as constant across the $k_\parallel$ range where the weight lives: it gives every line-of-sight mode the power of the $k_\parallel = 0$ mode, $P(k_\perp)$. The power then leaves the integral, and Eq. (10) finishes the calculation:

$$C^{ij}_{\rm Limber}(\ell) \propto P(k_\perp) \int_{-\infty}^{\infty} \frac{dk_\parallel}{2\pi} \,\mathrm{Re}\big[W_i W_j^*\big] = P(k_\perp) \int d\chi \, n_i(\chi)\, n_j(\chi). \tag{11}$$

This is the flat-sky version of Eq. (1): the cross spectrum collapses to the kernel overlap times the power. For two separated shells it is zero at every ℓ – not because no mode connects the shells, but because every mode is given the same power and the balanced weights then cancel exactly. The overlap O_{ij} of Eq. (4) is therefore a Limber-level quantity: it decides the Limber cross spectrum, not the exact one.

The exact integral does not cancel. Away from the approximation each mode carries its own power, $P(\sqrt{k_\perp^2 + k_\parallel^2})$. Panel (b) of Figure 4 draws that power over the single power Limber assigns, at two multipoles, with the $k_\parallel$ of negative weight shaded. At $\ell = 20$ the ratio rises to about 1.27 inside the first shaded band: the anticorrelated modes carry about a quarter more power than the correlated ones, the cancellation fails, and the sum comes out negative. At $\ell = 200$ the ratio stays close to one across the first bands, the cancellation nearly succeeds, and the exact cross spectrum is tiny: there Limber is accurate even for separated shells.

The scale below which the shells correlate. The weight of Eq. (9) lives at $k_\parallel$ up to a few times $1/d$. The power varies appreciably across that range only while $k_\perp$ is not much larger than $1/d$, that is, for

$$\ell \lesssim \frac{\chi}{d}. \tag{12}$$

At $\chi = 3200$ Mpc with shells of $w \approx 55$ Mpc: bins two apart have $d = 2w = 110$ Mpc, so in real space they correlate below $\ell \approx 29$; bins three apart have $d = 165$ Mpc and $\ell \approx 19$. Touching neighbours have $d = w = 55$ Mpc and $\ell \approx 58$ – and they also hold galaxy pairs at arbitrarily small separation, which carry their correlation far beyond that scale (Table 5).

Why the low-ℓ sign is negative. At $\ell = 20$ the transverse wavenumber is $k_\perp = \ell/\chi \approx 0.006$ Mpc^{-1}, below the peak of the power spectrum (near $k = 0.013$ Mpc^{-1} in the toy's BBKS fit). Adding a line-of-sight component moves a mode toward the peak, so it gains power, and the modes that gain most within the first lobes of the weight sit near $k_\parallel = \pi/d$: the opposite-sign modes. The negative weights win, so the cross spectrum of separated thin shells is negative at low ℓ – the connecting modes have about twice the centre separation for a wavelength, a crest in one shell and a trough in the other. Panel (c) carries the numbers, as the correlation coefficient $r_\ell = C^{ij}_\ell / \sqrt{C^{ii}_\ell C^{jj}_\ell}$ of Section 5.5: in real space the shells two apart have $r_\ell = -0.031$ at $\ell = 20$, cross zero near $\ell = 40$ and reach $+0.008$ at $\ell = 100$; the touching shells hold $r_\ell \approx +0.19$ at both multipoles.

What redshift-space distortions add. The bins are cut in measured redshift, and the coherent infall of galaxies toward large structures shifts those redshifts: the Kaiser distortion

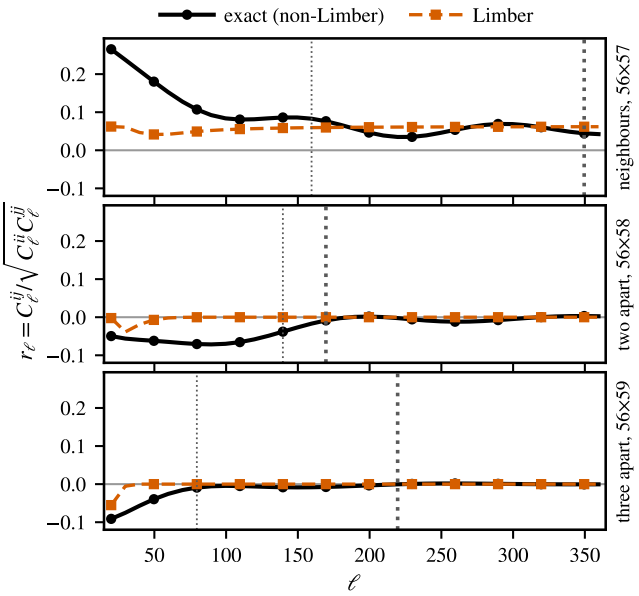


Figure 5: The correlation coefficient r_ℓ of $\Delta z = 0.02$ bin 56 with bin 57 (neighbours), 58 (two apart) and 59 (three apart): the forecast's exact non-Limber spectra (black solid, circles) and the Limber approximation of the same model (vermillion dashed, squares), signal only; gray: zero. Dotted vertical lines: the last band the UV-fraction cut keeps at $k_{\max} = 0.1$ (thin) and $0.2\,h$/Mpc (thick). $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

of Section 5. In linear theory [6] the power of a mode becomes

$$P_s(k_\perp, k_\parallel) = (b + f\mu^2)^2\, P(k), \qquad \mu = \frac{k_\parallel}{k}, \tag{13}$$

with b the linear bias, $f = d\ln D/d\ln a$ the growth rate (a the scale factor) and μ the cosine between the mode and the line of sight; the toy takes $b = 1.5$ and $f = 0.8$, roughly an ELG bin at $z \approx 0.9$. The power now depends on direction: $\mu^2 = k_\parallel^2/(k_\perp^2 + k_\parallel^2)$ varies across $k_\parallel$ on a scale set by $k_\perp$ itself, whatever the gap, so the Limber cancellation fails up to much higher ℓ. The physical picture matches: infall moves galaxies across bin edges in redshift, so the velocities themselves couple separated shells. The dashed curves of panel (c) show it: the shells two apart reach $r_\ell = -0.107$ at $\ell = 20$ and still -0.038 at $\ell = 100$, three times the real-space value at the lowest multipoles and persisting where the real-space curve has already crossed zero.

A measured pair of the forecast. Figure 5 repeats the comparison on the forecast itself, with nothing idealized. The bins are 56 to 59 of $\Delta z = 0.02$ (ELG1, $\bar{z} = 0.930$, 0.950, 0.970 and 0.990). The exact curves are the fiducial data vector of the layout's pair-axes derivative base (Section 12), computed without the Limber approximation, with redshift-space distortions and the Fingers-of-God blur. The Limber curves are one evaluation of the same model with every clustering spectrum, auto and cross, in the Limber approximation. Both are the signal alone; a cross spectrum has no shot noise anyway (Section 4). Table 5 compares the lowest band.

For the neighbours the exact coefficient at the lowest band is 0.265 against Limber's 0.06:

pair	bins	d [Mpc]	χ/d	exact r_ℓ	Limber r_ℓ
neighbours	56×57	55	58	$+0.265$	$+0.06$
two apart	56×58	110	29	-0.050	-0.002
three apart	56×59	165	19	-0.091	-0.055

Table 5: The correlation coefficient of the lowest band ($\ell = 20$) of $\Delta z = 0.02$ bin 56 with its three partners, exact and in the Limber approximation (Figure 5); d is the approximate comoving separation of the bin centres and χ/d the multipole scale of Eq. (12).

Limber underestimates the neighbour correlation by a factor of about four there, and only above $\ell \approx 200$ do the two settle together near 0.06, the high-ℓ value the blurred overlap of Section 5.5 predicts. For the pair two apart the exact coefficient is -0.05 to -0.07 from the lowest band out to $\ell \approx 110$ and is gone above $\ell \approx 170$; three apart it starts at -0.091 and is gone above $\ell \approx 80$. The two-apart correlation persists well beyond the real-space scale $\chi/d \approx 29$, exactly like the toy's Kaiser curves: redshift-space distortions carry it there.

Why the Limber curves are not exactly zero. The Limber curves of the two separated pairs are not quite zero at the lowest bands, although Eq. (11) says they should be. The density term is: with no kernel overlap it vanishes. What remains is the redshift-space term, which the code evaluates in an extended-Limber form whose kernel samples each bin's $n(z)$ at two line-of-sight points, $\chi = \ell/k$ and $\chi' = (\ell+1)/k$, a distance χ/ℓ apart (`W_RSD` in `cosmolike/radial_weights.c`). At $z \approx 0.95$ that distance is about 110 Mpc at $\ell \approx 30$ and 165 Mpc at $\ell \approx 20$: just the separations of the bins two and three apart. So at the lowest multipoles even the Limber redshift-space term couples non-overlapping bins; as ℓ grows, χ/ℓ shrinks below the gap and the Limber curves drop to zero.

The kept bands sit where the signal lives. The dotted lines of Figure 5 mark the scale cut of the forecasts (Section 11). At $k_{\max} = 0.1\,h$/Mpc the cut keeps 13 bands of the two-apart pair, out to $\ell = 140$, and 7 of the three-apart pair, out to $\ell = 80$; at $0.2\,h$/Mpc, 16 (to 170) and 21 (to 220). Those are the multipoles where the exact correlation lives. When the larger pair lists of Section 12 add such a pair, the bands that survive the cut are the low-ℓ bands that carry its non-Limber signal.

What this means for the rest of the note. The pair rule selects by kernel overlap, a Limber-level criterion; the forecast is non-Limber, so the pairs the rule excludes still carry a few percent of correlation at the lowest bands. And that correlation is a special measurement: the cross spectrum scales as $b_i b_j$ and the two autos as b_i^2 and b_j^2, so the linear biases divide out of r_ℓ at linear order. The larger pair lists of Section 12 add exactly this bias-free record of how clustering continues from shell to shell, which is why their gain flows through the free bias levels and slopes (Section 13). The analysis keeps the current list; the larger lists are a comparison against it.

What to remember.

- The Limber approximation gives every line-of-sight mode the power of the purely transverse one. Without kernel overlap the mode weights then cancel exactly, so Limber

predicts a zero cross spectrum for separated bins at every ℓ: the overlap O_{ij} is a Limber-level criterion.

- The exact projection weights each mode by its own power, so the cancellation fails at $\ell \lesssim \chi/d$; redshift-space distortions carry the correlation to higher multipoles still.
- At the lowest bands the correlation of separated thin shells is negative and of a few percent: the connecting modes put a crest in one shell and a trough in the other, and at these scales those modes carry the most power.
- The linear biases divide out of the correlation coefficient of two separated bins, so the pairs outside the standard list measure clustering growth with no bias in the way (Sections 12 and 13).

10 What is not measured, and why that is a choice

Unmeasured spectra are not zero. Two thin bins one apart share no galaxies and, even blurred, no redshift range, so their cross spectrum vanishes in the Limber regime. At low ℓ it does not vanish: modes longer than a bin along the line of sight connect the two shells (Section 9), so their non-Limber cross spectrum is small but nonzero – a few percent of correlation at the lowest bands, negative, gone above $\ell \approx \chi/d$ for shells a distance d apart. The same holds for any pair beyond. After the second step of the rule no unmeasured pair of any layout has overlapping blurred kernels, so every unmeasured spectrum vanishes in the Limber regime and lives at low ℓ only.

Leaving a spectrum out is a choice of data, not an error. The likelihood compares each spectrum in the data vector with the model's prediction for that same spectrum, weighted by the covariance of the measured spectra. Leaving a pair out removes its rows from the data, the model and the covariance together. The remaining comparison is still a correct one; it uses less information. So the choice can cost precision, by the information the unmeasured spectra carry, but it cannot bias a fit: a bias would need a wrong prediction for a spectrum that is kept, and leaving another spectrum out changes no prediction. How much precision the current list gives up is measured in Section 12.

11 How much do the low-overlap pairs carry?

The test. How much information do the weakly overlapping pairs contribute? The test keeps a cross spectrum only when its blurred overlap at the default dispersions reaches a floor, $O_{ij} \geq O_{\min}$ (Eq. (4)), applied to *every* measured pair, the DESI-imported ones included; the auto spectra are always kept, a kept pair keeps its bands, and the covariance is rebuilt on the kept rows. The fit is the fiducial model on the pessimistic linear-bias prior, with α_2 free and the exploration σ_v widths (Section 2): the reference model of `notes/fog_parameter_degeneracy_study`. It is scored by the dark-energy figure of merit $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$, the inverse area of the marginalized ellipse of w_0 and w_a (the dark-energy equation of state $w(a) = w_0 + w_a(1 - a)$, a the scale factor), and by $\mathrm{FoM}(\Omega_m, 10^9 A_s)$, defined the same way for the matter density Ω_m and the primordial amplitude A_s. Marginalized means that every

other parameter is left free, so its uncertainty widens the ellipse: $\mathrm{Cov}(w_0, w_a)$ is the (w_0, w_a) block of the inverse Fisher matrix (`notes/fisher_machinery`), and a larger figure of merit is a tighter dark-energy measurement. Each score is quoted over $\mathrm{FoM}_{\mathrm{current}}$, the same figure of merit of the current data vector: the pairs whose blurred overlap at the default dispersions reaches 10^{-6}, plus the BGS–LRG1 boundary pair of $\Delta d_{\mathrm{com}} = 62$ Mpc (Section 6). The fit carries the forecast's scale cut, computed on the paper model's fiducial (Section 2): bin i keeps wavenumbers below $k_{\mathrm{max}}/D(\bar{z}_i)$, with D the linear growth factor ($D(0) = 1$) and k_{max} always quoted at $z = 0$; Figure 6 shows $k_{\mathrm{max}} = 0.075$ and $0.2\,h/\mathrm{Mpc}$.

Which pairs drop. Every measured pair but one overlaps by at least 0.028 on $\Delta d_{\mathrm{com}} = 62$ Mpc, 0.043 on $\Delta z = 0.02$, 0.015 on $\Delta z = 0.05$ and 0.0034 on $\Delta z = 0.2$. The exception is the BGS–LRG1 boundary pair of $\Delta d_{\mathrm{com}} = 62$ Mpc, bins 18 and 19, whose blurred kernels overlap by 3.4×10^{-7} (Section 6). So the floors $O_{\mathrm{min}} = 10^{-6}$, 10^{-4} and 10^{-3} all drop that one pair and nothing else, and $O_{\mathrm{min}} = 10^{-2}$ additionally drops 10 of the 13 pairs of $\Delta z = 0.2$, whose overlaps all sit below 10^{-2} (Section 5.5).

What it costs. Dropping the boundary pair changes $\mathrm{FoM}(w_0, w_a)$ by a factor 0.999 at $k_{\mathrm{max}} = 0.2\,h/\mathrm{Mpc}$ and nothing at the smaller cuts – the only difference between the "current" point and the 10^{-6} floor in Figure 6 – and every other layout is unchanged at every floor up to 10^{-3}. At 10^{-2} the $\Delta z = 0.2$ figures of merit fall to 0.82 of $\mathrm{FoM}_{\mathrm{current}}$ for (w_0, w_a) and to 0.65–0.72 for $(\Omega_m, 10^9 A_s)$.

The reading. The only listed pair with overlap below 10^{-3} is the boundary pair, and it costs 0.1% of one figure of merit at the largest cut. The list holds no pair between 3.4×10^{-7} and 3.4×10^{-3} – on the layouts other than $\Delta z = 0.2$ the next-smallest overlap is 0.015–0.043 – so this test cannot say where between those overlaps a pair begins to matter. And a small overlap does not mean little information: the 13 pairs of $\Delta z = 0.2$ overlap by only 0.3–0.9%, yet removing ten of them costs up to 35% of $\mathrm{FoM}(\Omega_m, 10^9 A_s)$.

12 What would pairs beyond the list add?

The question, reversed. Section 11 removed pairs from the current list and found that no listed low-overlap pair carries much. This section asks the opposite question: how much would the data vector gain if it measured pairs *beyond* the current list? Two families of pair lists answer it; each contains the current list and grows along one axis.

The gap axis. The first family loosens the first step of the pair rule (Section 6). For a whole number g it measures, besides the current list, every pair of bins whose raw $n(z)$ ranges are at most g bin widths apart; g counts the gap between the two ranges in units of the layout's bin width, so $g = 0$ is the current list and each step in g adds the next chain of pairs, $C_\ell^{i,i+2}$ at $g = 1$, $C_\ell^{i,i+3}$ at $g = 2$, and so on, with the matching pairs across the sample boundaries. On $\Delta d_{\mathrm{com}} = 62$ Mpc the list grows from 107 pairs to 190, 285 and 378 at $g = 1$, 2 and 3; the other layouts grow in the same way. Every added pair has zero kernel overlap, raw or blurred, so its whole signal is the low-ℓ non-Limber correlation of Section 9.

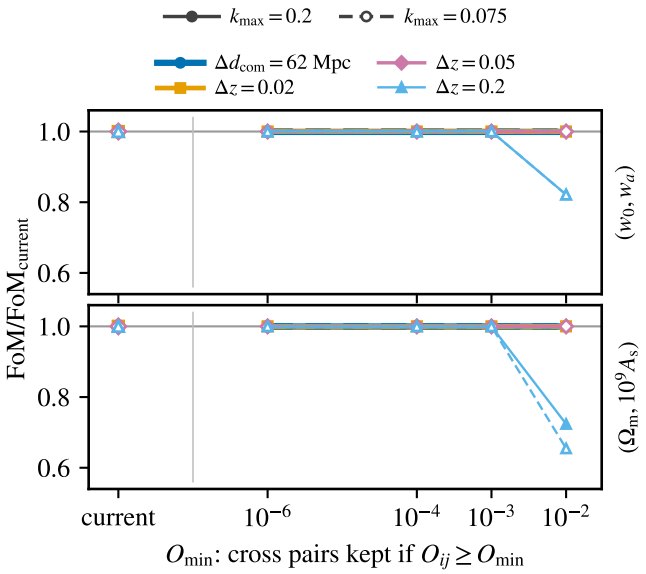


Figure 6: The figures of merit of the fit of Section 11 with only the cross pairs whose blurred overlap at the default dispersions reaches $O_{\min}$, over $\mathrm{FoM}_{\mathrm{current}}$, the figure of merit of the current data vector, against $O_{\min}$; "current", the current data vector with every measured pair, sits left of the separator. Upper panel: $\mathrm{FoM}(w_0, w_a)$; lower: $\mathrm{FoM}(\Omega_m, 10^9 A_s)$; the panels share the y axis. Color, line width and marker: the layout; solid, filled markers: $k_{\max} = 0.2\,h/\mathrm{Mpc}$; dashed, open: 0.075; gray: one. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

The σ_v axis. The second family loosens the second step. It reruns the blurred-overlap criterion, $O_{ij} \geq 10^{-6}$ (Eq. (4)), with every velocity dispersion set to one common value s in place of the seven defaults. The dispersion s only *selects* the pairs: the model and the covariance of every forecast keep the fiducial σ_v, so this axis measures what a longer list is worth, not what a stronger blur does. The lists change late: on $\Delta d_{\mathrm{com}} = 62$ Mpc no pair is added until $s = 750$ km/s, where the bins two apart begin to overlap (199 pairs); on $\Delta z = 0.02$ the list grows at every step, from 147 pairs at $s = 450$ to 344 at 1200; on $\Delta z = 0.05$ the first new pairs appear only at $s = 1200$, and on $\Delta z = 0.2$ none ever do. Among these lists are the ones that building the pair rule at the edge of the σ_v prior would produce (Section 14).

How it was computed. For each layout the derivative base – the change of every band of every spectrum per unit change of every model parameter, the input of a Fisher matrix – was computed once, on the layout's largest list (the union of every point's pairs: 378, 370, 178 and 32 cross pairs). Each point of Figures 7 and 8 is then a row selection: it keeps the rows of the data vector that belong to its own list (every auto band and the bands of the list's cross pairs that pass the scale cut; the shear rows are masked in these forecasts, which therefore use the clustering spectra alone) together with the matching rows and columns of the covariance, and builds its Fisher matrix from those alone. For a Gaussian likelihood this is exact, not an approximation: selecting the rows equals rebuilding the data set on the smaller list. Two checks confirm the machinery: on the shared rows every derivative base equals the current caches bit for bit, and the $s = 600$ point of $\Delta z = 0.02$ reproduces, to every digit, a separate

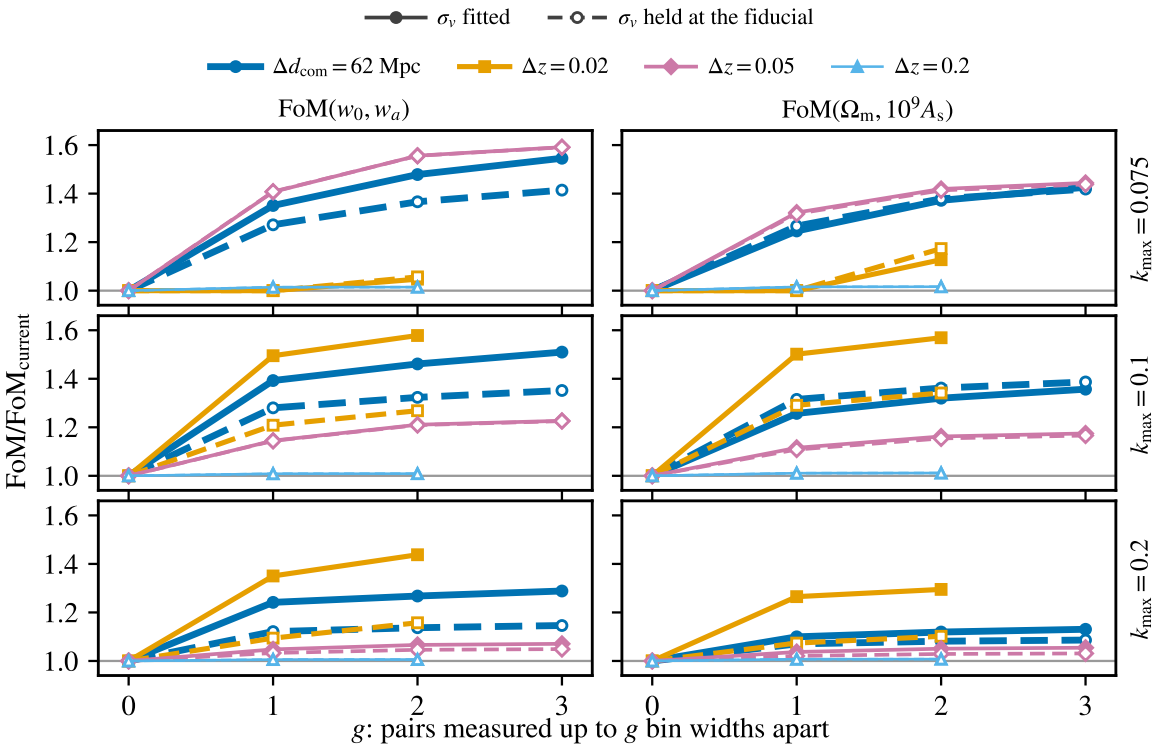


Figure 7: The figures of merit of the paper model when, besides the current list, every pair whose raw $n(z)$ ranges are at most g bin widths apart is measured, over FoM$_{\rm current}$ ($g = 0$, the current list). Rows: $k_{\rm max} = 0.075$, 0.1 and $0.2\,h$/Mpc; left column FoM(w_0, w_a), right FoM$(\Omega_m, 10^9 A_s)$; every panel shares one y range. Color, line width and marker: the layout; solid lines, filled markers: σ_v fitted; dashed, open: σ_v held at the fiducial; gray: one. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

forecast whose data vector, covariance and derivatives were built on exactly those pairs.

The model is the paper model (Section 2): the fiducial model, with α_2 fixed at zero in every bin and every σ_v prior width six times the exploration width, on the pessimistic linear-bias prior. Each list is forecast twice: once with the seven σ_v fitted – left free, so the quoted errors carry their uncertainty – and once with them held at the fiducial values. The reference model of Figure 6 behaves similarly; its grids are in the two figure folders' `data.json`.

What the figures show. No ratio can fall below one: the larger list contains the current one, and giving a Fisher forecast more data can only add information, never remove it. The question is how much is added. On the three thinner layouts the gain is large – at its largest point each layout gains 55–59% of FoM(w_0, w_a) – and the first added chain of pairs carries most of it. On $\Delta z = 0.2$ the gain never exceeds 2%. Two kinds of flat stretch in the figures mean selection, not physics: on the σ_v axis a ratio of one before the first jump only says that the list has not changed yet, and at $k_{\rm max} = 0.075\,h$/Mpc on $\Delta z = 0.02$ the scale cut removes every band of the added pairs up to $s = 900$ km/s (and at $g = 1$), so those points sit at one by the cut.

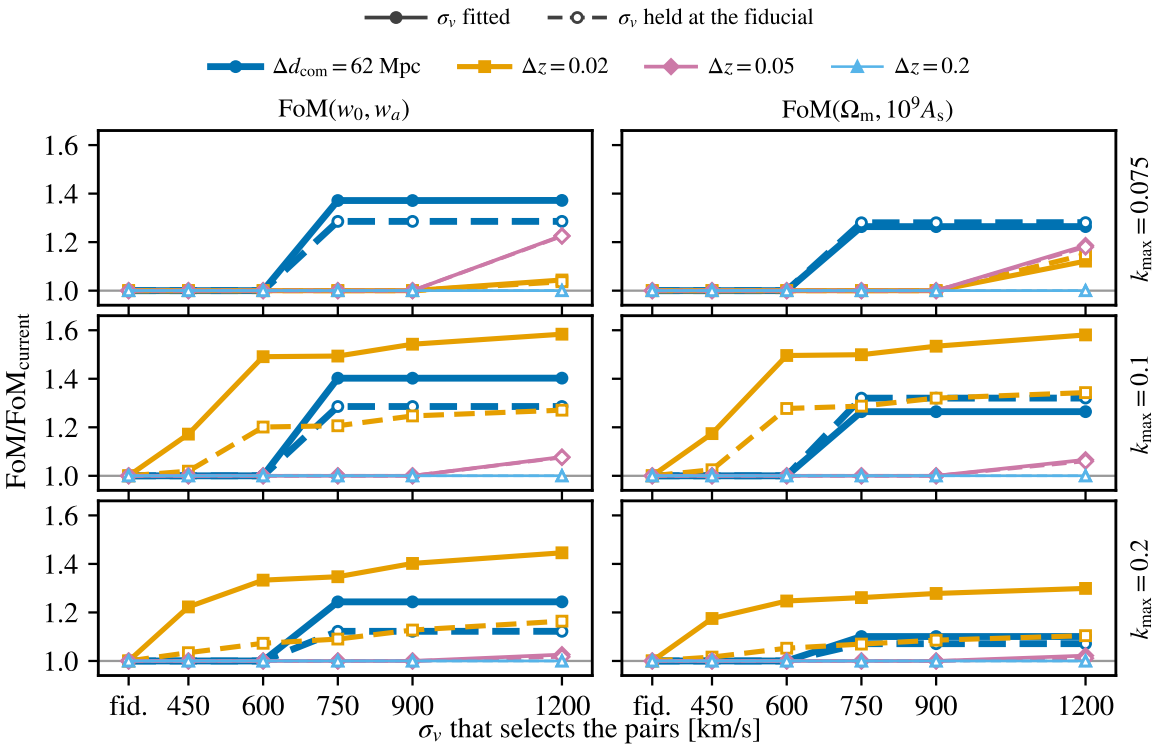


Figure 8: The figures of merit of the paper model when the pair list is selected with every velocity dispersion set to the common value on the horizontal axis, over FoM$_{\rm current}$ ("fid.", the selection at the default dispersions, is the current list). The dispersion only selects the pairs; the model keeps the fiducial σ_v. Rows: $k_{\rm max} = 0.075$, 0.1 and $0.2\,h$/Mpc; left column FoM(w_0, w_a), right FoM($\Omega_m, 10^9 A_s$); every panel shares one y range. Color, line width and marker: the layout; solid lines, filled markers: σ_v fitted; dashed, open: σ_v held at the fiducial; gray: one. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

Thin bins against wide bins. Every ratio so far compares a layout to its own current list. That answers what the added pairs are worth, but it hides the comparison the paper turns on: how much the thin slicing itself is worth. Figure 9 therefore redraws the gap axis against one common reference, the wide layout: every figure of merit is divided by that of $\Delta z = 0.2$ on its current list at the same scale cut. ($\Delta z = 0.2$'s own gain along the axis is at most 2%, so its ratio would stay within 2% of one; it is not drawn.) Read this way, the thin bins measure dark energy far better than the wide bins: on the current lists at $k_{\rm max} = 0.2\,h$/Mpc, FoM(w_0, w_a) is 26 times the wide layout's on $\Delta z = 0.02$ and 22 times on $\Delta d_{\rm com} = 62$ Mpc, against 8 on $\Delta z = 0.05$, and the added pairs raise the three factors to 37, 28 and 8. The advantage grows with the cut: at $0.075\,h$/Mpc the current lists hold factors of 8, 13 and 4.

One flat stretch needs the same care as before. On $\Delta z = 0.02$ at $k_{\rm max} = 0.075\,h$/Mpc the scale cut removes every band of the added pairs, so the curve barely moves along the axis: the factor that point holds over the wide layout is a property of the cut, not of the pairs.

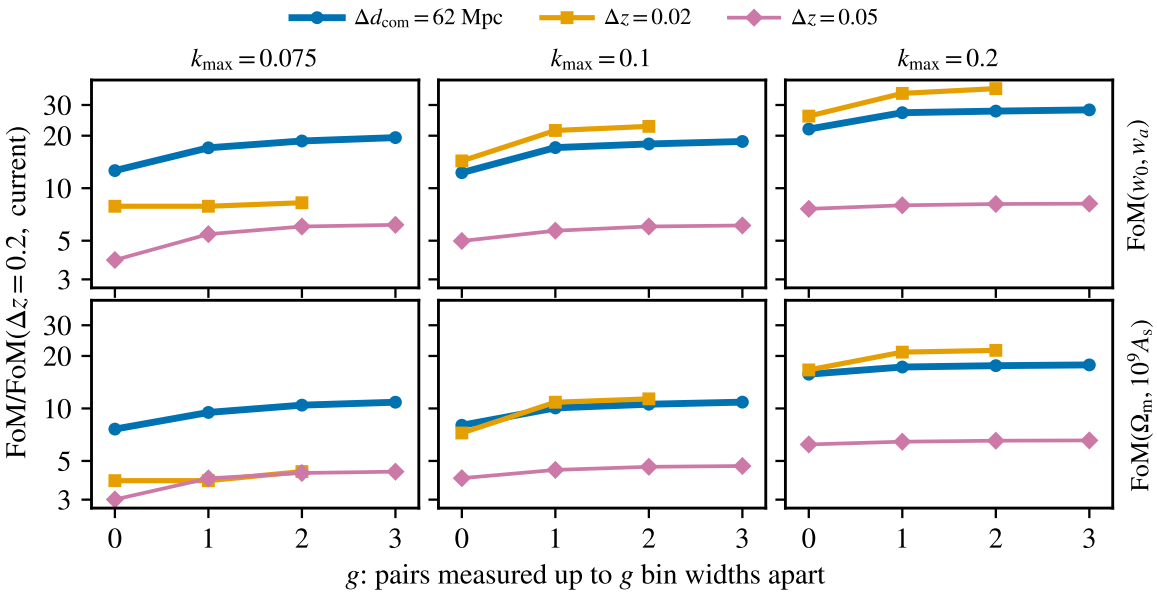


Figure 9: The figures of merit of the paper model along the gap axis, over those of the $\Delta z = 0.2$ layout on its current list at the same scale cut, with σ_v fitted ($\Delta z = 0.2$'s own gain along the axis is at most 2% and its ratio is not drawn). Rows: FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$; columns: $k_{\rm max} = 0.075$, 0.1 and 0.2 h/Mpc; every panel shares one logarithmic y range. Color, line width and marker: the layout. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

Why the figures of merit rise. The solid and the dashed curves separate two cases: the dashed curves hold the seven dispersions at their fiducial values, the solid curves fit them, and on the two thin layouts the solid curves gain visibly more. Section 13 traces both gains to one source. The auto and cross spectra of narrow bins together recover the three-dimensional clustering information, redshift-space distortions included [7]; in this model that information pays off mainly by separating each group's free bias level and slope from the growth history, and the dispersions ride on the same separation.

The neglected magnification. Every forecast here sets the magnification bias $b_{\rm mag}$ – the response of the observed number density to the lensing convergence – to zero, in the model and in the covariance. Lensing by the matter in front of a bin changes how many of its galaxies pass the flux limit, and for two bins with no kernel overlap that lensing is a correlated signal the model then misses. `notes/lensing_magnification` measures what the neglect costs as linear-response shifts of the fits, in units of each parameter's marginalized error ((w_0, w_a) against its error ellipse). The amplitude is one coefficient per sample, $C = 5s - 2$, with s the count slope, the logarithmic slope of the cumulative number counts at the flux limit; the measured slopes give $C = 2$ for BGS and 3 for LRG, and the unmeasured ELG slope is taken as $C_{\rm ELG} = 0$ with corners at ± 2.

At the measured slopes the shifts stay below about 0.07 on the three thinner layouts, on the current and the largest lists alike: magnification at gaps of a few thin-bin widths is still small, and the free bias levels and slopes absorb most of it. The ELG corners dominate the

budget: at $C_{\rm ELG} = +2$ the largest lists reach 0.30–0.46 in (w_0, w_a) against at most 0.06 on the current lists, and at -2 the current lists themselves reach 0.2–0.3 across the layouts. On $\Delta z = 0.2$ the measured cross spectra are magnification-dominated – two wide adjacent bins share little density signal while the lensing across a wide bin is strong, and magnification lifts one measured adjacent cross spectrum to 25 times its magnification-free size – so (w_0, w_a) shifts by 0.19 even at the measured slopes.

13 Why the added pairs help: the free bias levels and slopes

The question. The gains of Section 12 call for an explanation. The added pairs raise FoM(w_0, w_a) by up to 55–59% on the three thinner layouts, and on the two thin layouts they raise it more when the dispersions σ_v are fitted than when they are held fixed – although the added spectra barely respond to σ_v at all (Table 6). Which parameters turn the extra spectra into so much dark-energy precision? This section answers in two steps: it first measures how σ_v changes each kind of spectrum and which other parameters change it the same way, then refits every pair list with blocks of parameters held fixed and reads off which block the gain flows through.

Cosmology and nuisance parameters. The model's parameters are of two kinds. The cosmological parameters – A_s, n_s, H_0, Ω_b, Ω_m, w_0 and w_a – are the goal. The rest are nuisance parameters: parameters that describe the galaxies rather than the universe, but that the fit must determine from the same data. In the paper model they are the linear bias $b_{1,i}$ of every bin, fitted through its amplitude $\beta_i = b_{1,i}\, D(\bar{z}_i)$ – the bias times the growth factor, constant in redshift for a population that is conserved – under the pessimistic prior of Section 2: within each sample group the amplitudes may take any overall level and any slope in redshift without penalty, while each bin's departure from that straight line, its residual, carries a Gaussian prior of width 3% of the group's fiducial amplitude, correlated between bins as $\exp(-|z_i - z_j|/L)$ with $L = 0.05$. Per sample come the higher-order biases b_2 and b_{s^2} – the galaxies' responses to the squared density and to the tidal field – the counterterm α_0, their redshift tilts, and the two stochastic amplitudes SN_0 and SN_2 of Section 4 (the "shot noise" block below); and there are the seven σ_v. Every figure of merit of Section 12 is marginalized: it quotes the dark-energy errors with every one of these left free, so that their uncertainty is included in the error bars.

Templates, and what a degeneracy is. For a parameter p, its template (or response vector) is the derivative of every kept band of the data vector with respect to p, each divided by that band's error bar, the square root of that band's covariance diagonal: $\sigma_\ell^{-1}\, \partial C_\ell / \partial p$. The template is the pattern the parameter imprints on the data, in units of what the data can resolve, and a fit can tell two parameters apart only as far as their patterns differ. Treating each template as an arrow in a space with one axis per band, the squared length of the arrow is the information the bands carry on that parameter alone, and the cosine of the angle between two arrows says how alike two parameters act: ± 1 is the same pattern up to sign, 0 is two independent patterns. Two parameters whose templates are nearly parallel are degenerate: they change the predicted spectra in nearly the same pattern, so the fit measures only a combination of them and both error bars grow.

How each kind of spectrum responds to σ_v. Table 6 measures these quantities for σ_v and the spectra of $\Delta z = 0.02$, kind by kind.

spectra	bands kept	median $d\ln C_\ell/d\ln\sigma_v$	information on σ_v	$\cos(\sigma_v, b_1)$	$\cos(\sigma_v, w_0)$
autos C_ℓ^{ii}	1910	-0.086	3.6×10^3	-0.991	$+0.981$
neighbours $C_\ell^{i,i+1}$	1514	$+0.293$	1.3×10^3	$+0.932$	$+0.271$
two apart $C_\ell^{i,i+2}$	416	-0.042	1.2	-0.118	$+0.649$
three apart $C_\ell^{i,i+3}$	523	-0.002	0.13	-0.271	$+0.087$

Table 6: How the within-sample spectra of $\Delta z = 0.02$ (both bins from one galaxy sample) respond to σ_v at $k_{\max} = 0.1\,h/\mathrm{Mpc}$ (quoted at $z = 0$; each bin cut at $k_{\max}/D(\bar{z}_i)$), measured on the layout's largest derivative base with the diagonal of the covariance, moving every dispersion together. "Information": the squared length of the response vector defined in the text, the derivative of every kept band over its error bar; cos: the cosine between the σ_v response and the named parameter's. The neighbours are in the current list; the pairs two and three apart enter at $g = 1$ and 2.

Three facts in the table carry the section.

1. In the auto spectra, the σ_v template is almost exactly minus the bias template and almost exactly the w_0 template (cosines -0.991 and $+0.981$): blurring the kernels lowers every auto spectrum the way a lower bias or a changed w_0 does. Fitting σ_v is therefore expensive exactly where the bins are thin against the blur (Table 7): on the current list it costs a factor 2.7 of $\mathrm{FoM}(w_0, w_a)$ on $\Delta z = 0.02$ at $k_{\max} = 0.1\,h/\mathrm{Mpc}$ and 2.1 on $\Delta d_{\mathrm{com}} = 62$ Mpc, against 1.04–1.13 on $\Delta z = 0.05$ and nothing on $\Delta z = 0.2$.

2. The neighbour cross spectra, already in the current list, respond to σ_v with the opposite sign to the autos: the blur moves galaxies across the shared bin edge and so raises the neighbour correlation while it lowers the autos – a pattern no bias change mimics. On the current list they are what limits the degeneracy.

3. The pairs two and three bins apart barely respond to σ_v: their information on σ_v is 1.2 and 0.13, against 3.6×10^3 in the autos, with median responses near zero. The added pairs do not measure σ_v; they measure bias and cosmology with negligible σ_v dependence.

layout	$k_{\max} = 0.075$	0.1	0.2
$\Delta d_{\mathrm{com}} = 62$ Mpc	1.98 / 1.81	2.12 / 1.90	1.65 / 1.47
$\Delta z = 0.02$	2.22 / 2.24	2.72 / 2.19	1.78 / 1.43
$\Delta z = 0.05$	1.05 / 1.05	1.04 / 1.04	1.13 / 1.10
$\Delta z = 0.2$	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00

Table 7: The cost of fitting σ_v in the paper model: $\mathrm{FoM}(w_0, w_a)$ with σ_v fixed over the same figure of merit with σ_v fitted, on the current list / on the largest list of the gap axis. $k_{\max}$ in h/Mpc, quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

Table 7 reads the same facts from the side of the added pairs: they lower the cost of fitting σ_v on the two thin layouts and leave it unchanged where it was already small.

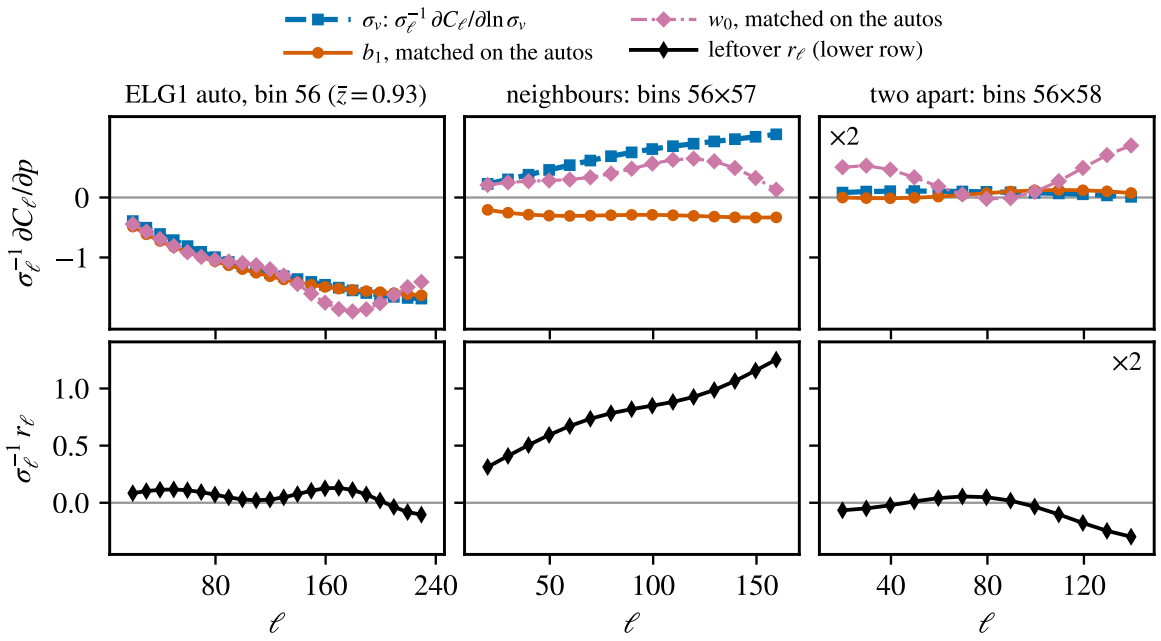


Figure 10: The σ_v, b_1 and w_0 templates of three $\Delta z = 0.02$ spectra of ELG1 bin 56 ($\bar{z} = 0.93$) at $k_{\max} = 0.1\,h/\mathrm{Mpc}$: the bin's auto spectrum, its cross spectrum with the next bin (in the current list) and with the bin two apart (added at $g = 1$; drawn at twice its size). Top row: the templates $\sigma_\ell^{-1}\,\partial C_\ell/\partial p$, with the b_1 and w_0 templates scaled by the coefficient that best matches the σ_v template over the kept bands of all 90 auto spectra. Bottom row: the leftover of the joint $b_1 + w_0$ match, with the same coefficients in every column. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

The σ_v template, band by band. Figure 10 draws these facts for one bin. The construction is the one of the template figure of `notes/fog_parameter_degeneracy_study` (there for α_2 against σ_v): the b_1 and w_0 templates of the bin's sample are scaled by the coefficient that best matches the σ_v template over the kept bands of all 90 auto spectra, so every panel shows how far the auto-spectrum match carries over to the other spectra. In the auto panel the three curves coincide, and the leftover after the joint $b_1 + w_0$ match is about a tenth of an error bar per band: within the autos, σ_v is indistinguishable from a bias change plus a w_0 change. In the neighbour cross the blur raises the spectrum while the matched bias change lowers it, and the leftover reaches 1.2 error bars per band: this is the part of the σ_v pattern that the current list resolves. In the pair two apart neither σ_v nor the matched bias does much, and the leftover stays small. One limitation: the figure moves one b_1 per sample, every bin of the sample together, so it cannot show what the added pairs constrain bin by bin. That is what the attribution fits measure.

The attribution fits. Which parameters turn the added spectra into dark-energy precision? The test repeats the forecasts of the gap axis with one block of nuisance parameters held at its fiducial values, as if those parameters were known exactly. In a Fisher forecast this takes one deletion: remove the block's rows and columns from the full Fisher matrix – the data

term plus every prior, the correlated prior of the bias residuals included – and invert what remains. This is conditioning, and for a Gaussian likelihood it is exact; it is the mirror image of marginalizing, which inverts the full matrix first and then reads off a sub-block, keeping the deleted parameters' uncertainty in the error bars. The blocks are the ones of the nuisance list above: the linear biases b_1, held fixed as a whole (every group's level, slope and residuals together); b_2 and b_{s^2} with their tilts; α_0 with its tilt; the shot-noise amplitudes; and every nuisance parameter at once, with cosmology and σ_v staying free. The linear-bias block is also split in two. Per group, the amplitudes β_i of its bins are written as a straight line in redshift plus residuals: the line spans the two directions the prior leaves free, the level and the slope, and the residuals span the directions it charges. Each part is then held fixed with the other left free. Three checks anchor the machinery: with no block deleted the ratios equal those of Section 12 to every printed digit, deleting the σ_v block reproduces the σ_v-fixed variants, and with both parts of the split left free the figures of merit equal the every-parameter-free ones, so the change of basis loses nothing.

What the fits show. Figure 11 draws the gap axis again, block by block, and Table 8 collects the key numbers. With every nuisance parameter fixed the gain nearly vanishes: the largest list is worth 2–8% where it was worth 55–59%, and the σ_v-fixed gains collapse the same way, to 2–7%. The line-of-sight clustering information that the added spectra carry by themselves is therefore worth a few percent once the biases are known. Most of the gain enters through the linear-bias block alone: holding it fixed cuts $\times 1.59$ to $\times 1.07$ on $\Delta z = 0.05$, $\times 1.58$ to $\times 1.15$ on $\Delta z = 0.02$ and $\times 1.55$ to $\times 1.26$ on $\Delta d_{\rm com} = 62$ Mpc (each at the scale cut of Table 8). Holding b_2 and b_{s^2} fixed removes a smaller part, and the counterterm and the shot noise change little. The split of the linear-bias block decides the reading (Table 9): holding only the residuals fixed, with every level and slope still free, barely changes the gain, while holding only the levels and slopes fixed removes almost exactly as much as fixing the whole block. The gain is carried by the free level and slope of each group's bias amplitude, not by the correlated per-bin residuals.

	$\Delta d_{\rm com} = 62$ Mpc	$\Delta z = 0.02$	$\Delta z = 0.05$
block held fixed	$k_{\rm max} = 0.075$	$k_{\rm max} = 0.1$	$k_{\rm max} = 0.075$
none	1.546 / 1.426	1.578 / 1.569	1.591 / 1.444
all b_1	1.256 / 1.312	1.146 / 1.189	1.072 / 1.083
b_2, b_{s^2}	1.399 / 1.328	1.230 / 1.247	1.339 / 1.230
α_0	1.525 / 1.392	1.445 / 1.435	1.464 / 1.355
shot noise	1.458 / 1.348	1.459 / 1.432	1.563 / 1.417
every nuisance	1.084 / 1.132	1.035 / 1.037	1.024 / 1.031

Table 8: The gain of the gap axis' largest list over the current list with one block of nuisance parameters held fixed on both, FoM(w_0, w_a) / FoM$(\Omega_m, 10^9 A_s)$, with σ_v fitted. Each layout is quoted at the scale cut where its gain is largest ($k_{\rm max}$ in h/Mpc, quoted at $z = 0$; each bin cut at $k_{\rm max}/D(\bar{z}_i)$) and at its largest gap point: $g = 3$, or $g = 2$ on $\Delta z = 0.02$.

The attribution against wide bins. The same conditioning answers a second question: how much of the thin-over-wide advantage of Figure 9 flows through the bias line? Figure 12 draws the gap axis once more, now over $\Delta z = 0.2$ on its current list, with each block of

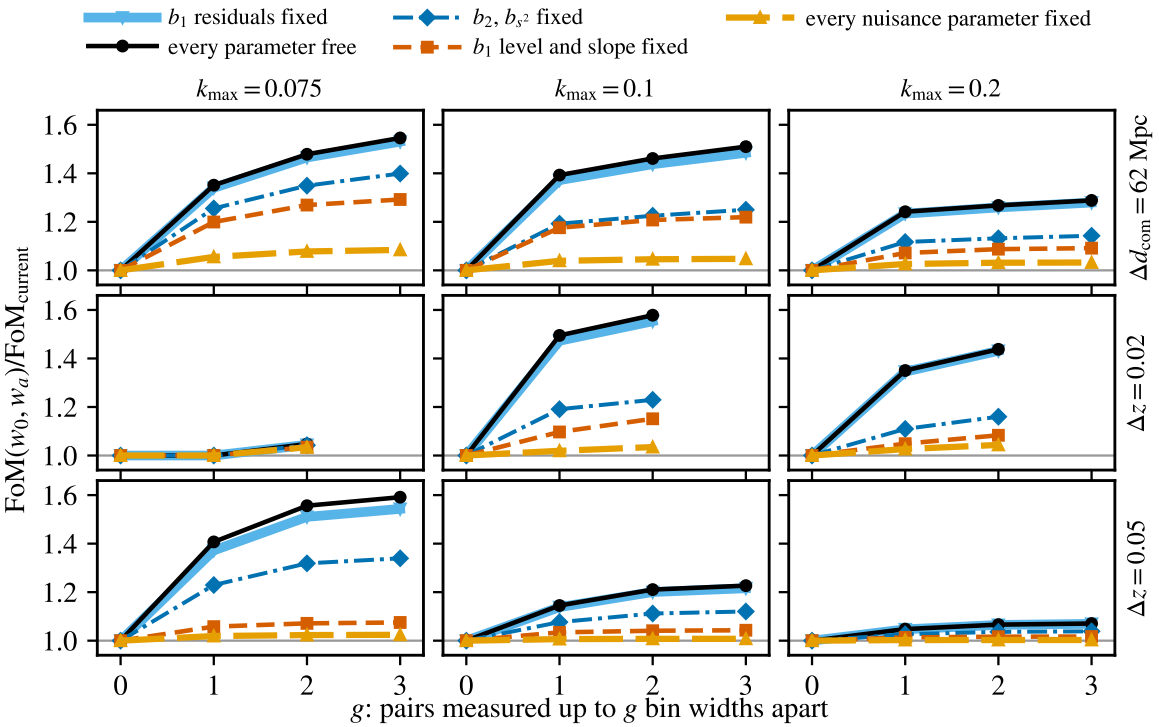


Figure 11: FoM(w_0, w_a) of the gap axis over the same figure of merit of the current list, with σ_v fitted and with blocks of nuisance parameters held at their fiducial values by conditioning; each curve holds the same block fixed on both lists. Rows: $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$ and $\Delta z = 0.05$ ($\Delta z = 0.2$ gains at most 2% and is not drawn); columns: $k_{\max} = 0.075$, 0.1 and $0.2\,h$/Mpc, quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$. Black solid: every parameter free; sky-blue wide solid (triangles), drawn under the black curve: the b_1 residuals fixed; blue dash-dotted: b_2 and b_{s^2} fixed; vermillion dashed (squares): the b_1 levels and slopes fixed; orange long-dashed: every nuisance parameter fixed.

nuisance parameters held fixed in the numerator and the denominator alike, so that both layouts are conditioned on the same knowledge and the ratio compares like with like. The answer is that most of the advantage comes from measuring the bias line. With each group's level and slope held fixed the thin layouts still beat the wide one, but by factors of about 7–9 at $k_{\max} = 0.2\,h$/Mpc instead of 22–26 (and by only 1.4–1.8 with every nuisance parameter fixed at 0.075). A thin layout, with many bins per sample group, measures the bias evolution across each group from its own spectra; the wide layout – two BGS, four LRG and four ELG bins, with sample groups of one or two bins – cannot, so freeing the bias line costs it far more. This is the same mechanism as the gain of the added pairs: separating the bias line from the growth history.

The figure also separates the two thin layouts. On the current lists $\Delta d_{\rm com} = 62$ Mpc leads $\Delta z = 0.02$ only at $k_{\max} = 0.075\,h$/Mpc (12.6 against 7.9 in FoM(w_0, w_a)); at 0.1 and 0.2 the order reverses. With the bias line fixed the two layouts are equal at 0.075 (4.4 against 4.6), so the $\Delta d_{\rm com} = 62$ Mpc lead at the conservative cut comes from constraining the bias line

layout, $k_{\max}$	all free	residuals fixed	level, slope fixed	all b_1 fixed
$\Delta z = 0.02$, 0.1	1.578	1.553	1.151	1.146
$\Delta z = 0.02$, 0.2	1.438	1.431	1.083	1.075
$\Delta z = 0.05$, 0.075	1.591	1.543	1.075	1.072
$\Delta d_{\rm com} = 62$ Mpc, 0.075	1.546	1.530	1.292	1.256
$\Delta d_{\rm com} = 62$ Mpc, 0.1	1.510	1.484	1.219	1.201
$\Delta d_{\rm com} = 62$ Mpc, 0.2	1.288	1.279	1.092	1.085

Table 9: The split of the linear-bias block: FoM(w_0, w_a) of the gap axis' largest list over the current list, with σ_v fitted and with parts of the block held fixed on both lists. "Residuals fixed" keeps every group's level and slope free; "level, slope fixed" keeps the residuals free. $k_{\max}$ in h/Mpc, quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

better there. Why it does has not been tested; its finer slicing at low redshift – 18 BGS bins of $\Delta z \approx 0.016$ against 15 bins of 0.02 – is a candidate. The two layouts slice essentially the same galaxy samples (per-sample densities within 2–3%, mean redshifts within 0.01), so the difference is in how the bins are cut, not in which galaxies are used.

The mechanism. On large scales the galaxy map of bin i is the matter map times its linear bias b_i (Section 4), and within each group the amplitude $\beta_i = b_i D(\bar{z}_i)$ is in effect a free straight line in redshift – the bias line of this section: the prior charges only the per-bin departures from that line. The auto spectra cannot pin the line down by themselves. A higher level raises every auto spectrum of the group through b_i^2 the way a larger clustering amplitude does, so the level is degenerate with A_s and the growth of structure; a slope in redshift tilts the group's spectra against each other the way an evolving growth does, and how growth evolves with redshift is what w_0 and w_a control. And σ_v, which in the autos acts like a lower bias, rides on the same degeneracy. A cross spectrum depends on the biases differently, as the product $b_i b_j$, and its ratio to the two auto spectra – the correlation coefficient $C_\ell^{ij}/\sqrt{C_\ell^{ii} C_\ell^{jj}}$ of Section 5.5 – does not depend on the linear biases at all at linear order (Section 9). Each added pair is therefore one more measurement of the clustering shared by two shells that no bias line can reshape: it separates the bias line from the growth history. This is also why the gain collapses when the levels and slopes are held fixed, and barely moves when only the residuals are: where the bias line is known, the auto spectra already measure what the added pairs would add.

Why the gain is larger with σ_v fitted. The added pairs carry almost no information on σ_v itself (Table 6), yet the σ_v-fitted curves gain more than the σ_v-fixed ones. The reason is that σ_v rides on the degeneracy between the bias line and growth: in the auto spectra it acts like a lower bias (Figure 10), so whatever separates the bias line from cosmology also separates σ_v. The attribution fits show the same link from the other side: with the bias levels and slopes held fixed, the cost of fitting σ_v on the *current* list falls from 2.7 to 1.9 on $\Delta z = 0.02$ at $k_{\max} = 0.1\,h$/Mpc, while holding only the residuals fixed leaves it at 2.7; with the whole bias block fixed it falls from 2.1 to 1.5 on $\Delta d_{\rm com} = 62$ Mpc. The extra gain of the fitted curves shrinks with the cost.

Magnification rides it too. The neglected magnification of Section 12 enters through the same freedom. At the measured count slopes the free levels and slopes absorb part of the

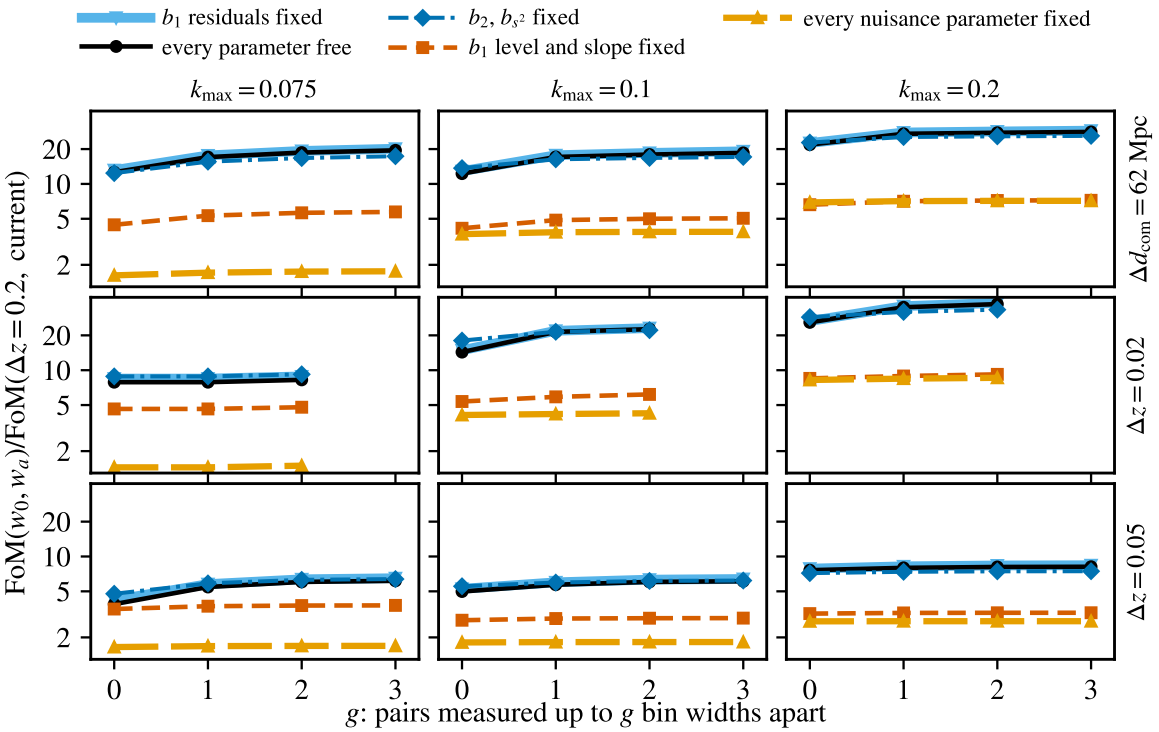


Figure 12: FoM(w_0, w_a) of the gap axis over the same figure of merit of the $\Delta z = 0.2$ layout on its current list, with σ_v fitted and with blocks of nuisance parameters held at their fiducial values by conditioning; each curve holds the same block fixed in the numerator and the denominator. Rows: $\Delta d_{com} = 62$ Mpc, $\Delta z = 0.02$ and $\Delta z = 0.05$; columns: $k_{max} = 0.075$, 0.1 and 0.2 h/Mpc, quoted at $z = 0$; each bin is cut at $k_{max}/D(\bar{z}_i)$; the y axis is logarithmic. Black solid (circles): every parameter free; sky-blue wide solid (triangles), drawn under the black curve: the b_1 residuals fixed; blue dash-dotted (diamonds): b_2 and b_{s^2} fixed; vermillion dashed (squares): the b_1 levels and slopes fixed; orange long-dashed (triangles): every nuisance parameter fixed.

neglected signal away from the cosmology: freezing every nuisance parameter raises the largest (w_0, w_a) shifts by a factor of 1.7 to 3.5 on the three thinner layouts. At the ELG corner $C_{ELG} = +2$ the balance flips on the largest lists: their magnification signal is read as a change of the bias lines and pulls the cosmology with it, so the free fits shift *more* than the frozen ones (0.46 against 0.20 on $\Delta z = 0.05$ at its largest list). The freedom that buys the gain of the added pairs is the freedom that lets an unmodeled magnification signal move the cosmology; the two travel together (`notes/lensing_magnification`).

13.1 How the gain depends on the linear-bias prior

The attribution fits all ran under one prior, the paper's pessimistic one. A natural worry is that the gain itself is an artifact of that choice. Three priors on the amplitudes β_i pass the accuracy test of `notes/correlated_bias_prior_v2`, which checks that the error bars a prior produces still cover the true cosmology across simulated universes:

- the baseline prior: per group a free level and a free slope in redshift, plus residuals of width 1% of the group's fiducial amplitude (3% for BGS), correlated as $\exp(-|z_i - z_j|/L)$ with $L = 0.2$;
- the pessimistic prior, the paper's choice: the same structure with 3% in every group and $L = 0.05$;
- the free prior: every β_i free, with no prior at all.

A fourth, the growth-tied prior – one amplitude per group, with no slope and no residuals – fails that accuracy test, so it is excluded and no figure of merit is quoted for it anywhere in this note.

Figure 13 repeats the gap axis under each passing prior, every point over the same prior's fit on the current list, and Table 10 gives the ratios at each layout's best-gaining scale cut. The three curves nearly coincide: the gain agrees within a few percent of the ratio across the priors, in the expected order – the looser the residual prior, the slightly larger the gain, because the added pairs have a little more left to determine. The absolute figures of merit do differ (the tighter the prior, the larger the figure of merit on either list), but the added pairs raise each by about the same factor. The agreement follows from the attribution: all three priors leave each group's level and slope of $\beta(z)$ completely free and differ only in the residual prior, and the residuals play almost no part in the gain (Table 9). What no passing prior does is constrain the levels and slopes themselves; the only tested prior without a free slope is the growth-tied one, which the accuracy test excludes.

	$\Delta d_{\rm com} = 62$ Mpc	$\Delta z = 0.02$	$\Delta z = 0.05$
prior	$k_{\rm max} = 0.075$	$k_{\rm max} = 0.1$	$k_{\rm max} = 0.075$
baseline	1.535 / 1.422	1.563 / 1.554	1.558 / 1.418
pessimistic (paper)	1.546 / 1.426	1.578 / 1.569	1.591 / 1.444
free	1.551 / 1.430	1.585 / 1.576	1.606 / 1.456

Table 10: The gain of the gap axis' largest list over the current list under each linear-bias prior that passes the accuracy test, FoM(w_0, w_a) / FoM$(\Omega_m, 10^9 A_s)$, with σ_v fitted and each ratio taken within one prior. Each layout is quoted at the scale cut where its gain is largest ($k_{\rm max}$ in h/Mpc, quoted at $z = 0$; each bin cut at $k_{\rm max}/D(\bar{z}_i)$) and at its largest gap point: $g = 3$, or $g = 2$ on $\Delta z = 0.02$.

What this establishes, and what it does not. The gain of Section 12 is a property of the bias model. It exists because each group's bias level and slope are completely free: the added pairs earn their keep by separating that bias line – and σ_v with it – from the growth history, while the correlated residuals play almost no part. It is not an artifact of the paper's pessimistic choice: every linear-bias prior that passes the accuracy test leaves the levels and slopes free and gives the same gain within a few percent (Section 13.1). What remains open is a finite prior on the levels and slopes themselves, which no passing prior imposes. A forecast with such prior knowledge would gain less from the added pairs, with the level-and-slope-fixed curves of Figure 11 as the limiting case, but none has been run: how much of the gain such a prior keeps is an open point (Section 14), not a result of this note.

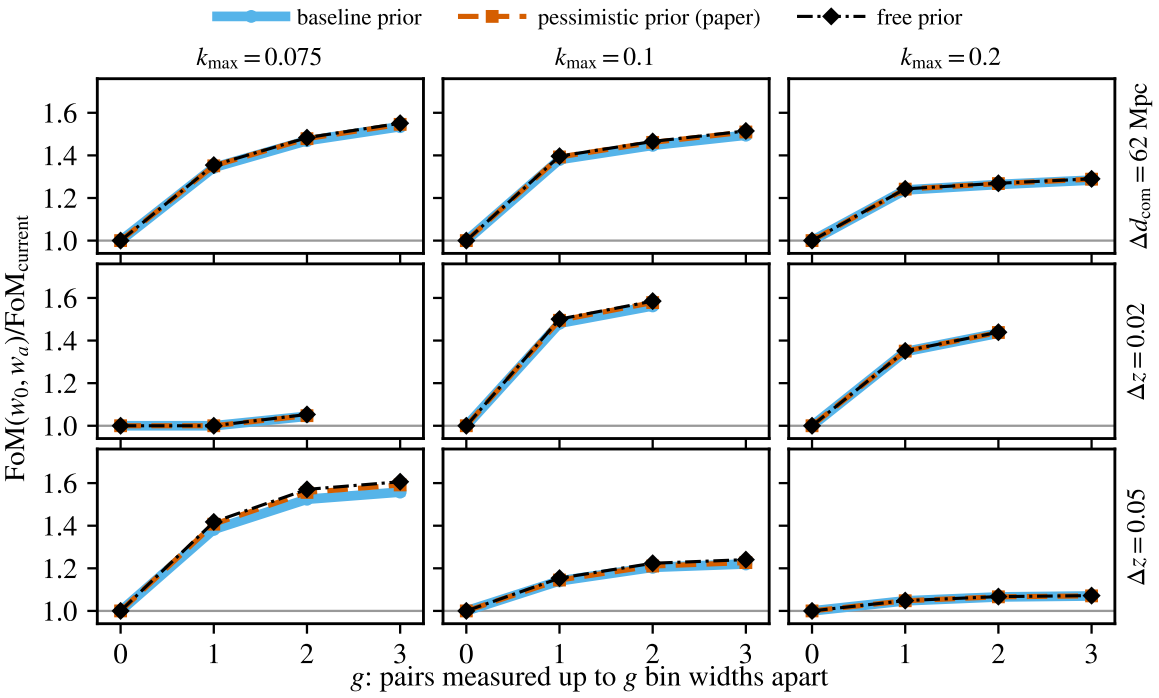


Figure 13: FoM(w_0, w_a) of the gap axis over the same figure of merit of the current list, with σ_v fitted, under each linear-bias prior that passes the accuracy test of `notes/correlated_bias_prior_v2`; each curve is normalized to its own prior's fit on the current list. Rows: $\Delta d_{com} = 62$ Mpc, $\Delta z = 0.02$ and $\Delta z = 0.05$ ($\Delta z = 0.2$ gains at most 2% and is not drawn); columns: $k_{max} = 0.075$, 0.1 and $0.2\,h$/Mpc, quoted at $z = 0$; each bin is cut at $k_{max}/D(\bar{z}_i)$. Sky-blue wide solid (circles): the baseline prior; vermillion dashed (squares): the pessimistic prior of the paper; black dash-dotted (diamonds): every amplitude free.

14 Open issues

- **The BGS–LRG1 boundary pair of $\Delta d_{com} = 62$ Mpc.** It is measured, as the DESI forecast measures it, although its ranges are 0.010 apart and its blurred kernels barely overlap (3×10^{-7}, below the second step's threshold); its signal is the low-ℓ non-Limber correlation alone (Section 9).

- **The list does not follow σ_v.** It is built at the default dispersions. A sampled σ_v changes every kernel overlap but not which pairs are measured. Within two prior widths of the defaults no unmeasured pair of any layout reaches an overlap of 10^{-4}; what the lists selected at larger dispersions would measure is forecast on the σ_v axis of Section 12.

- **The pair list at the σ_v prior edge.** Just beyond those two prior widths the thin layouts' bins two apart begin to overlap by more than 10^{-6} (`notes/fog_parameter_degeneracy_study`, on the pair rule at the edges of the σ_v prior). Those lists have now been forecast (Section 12, the σ_v axis), and the gain is attributed to the free level and slope of each group's bias amplitude (Section 13), and the largest lists' derivative bases

are archived in `fisher_caches/`. The analysis keeps the current list; the larger lists are a comparison against it.

- **The gain under a prior on the bias levels and slopes.** Every linear-bias prior that passes the accuracy test leaves each group's level and slope free, and all of them give the same gain within a few percent (Section 13.1); with the levels and slopes exactly fixed the gain falls from 55–59% to 8–29% (Section 13). A finite prior on them would land between those curves; no forecast with one has been run, so where the gain lands is not known.
- **The neglected magnification.** Every forecast sets the magnification bias $b_{\rm mag}$ to zero. At the measured count slopes that costs at most about 0.07 of the marginalized errors on the three thinner layouts, on the current and the largest lists alike, but the unmeasured ELG count slope dominates the budget: at its $C_{\rm ELG} = +2$ corner the largest lists shift by 0.30–0.46 in (w_0, w_a) (Section 12; `notes/lensing_magnification`). Measuring the ELG count slope, and putting magnification into the model, are the open steps.

Reproducing the figures. `plots/make_cross_correlations_data.py` reads the project's `data/` and `likelihood/` files and writes the `data.json` of the first three figure folders, with every number quoted for them here; `plots/pair_threshold/data.json` is written by the Fisher notebook `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb` (section 11, item 7). The pair-axes figures start from the archived derivative, template and scale-cut caches of each layout's maximal pair list, `model_{derivatives,templates,scale_cuts}_<layout>_pairmax_final.npz` in `fisher_caches/` (`<layout>` = `dcom62`, `dz0p02`, `dz0p05`, `dz0p20`), each copied to `chains/` with an explicit `fm.load_fisher_from_cache` call, and from each layout's pair-list covariance, built with the project's covariance command on the `EXAMPLE_EVALUATE_COVARIANCE*_PAIRMAX.yaml` examples. `scripts/compute_pair_axes_fits.py --layout <layout>` then writes `chains/pair_axes_fits_<layout>.json`, the input of the five generators of `plots/`: `make_pair_axes_data.py`, `make_pair_axes_attribution_data.py`, `make_pair_axes_priors_data.py`, `make_pair_axes_wide_ratio_data.py` and `make_pair_axes_attribution_wide_data.py`. The docstrings of the script and the generators state the exact prerequisites and arguments. `plots/make_sv_templates_data.py` writes `plots/sv_templates`' `data.json` from the $\Delta z = 0.02$ pairmax derivative base and data set files. `plots/make_limber_toy_data.py` writes `plots/limber_toy`'s `data.json` with numpy alone, and `plots/make_limber_real_data.py` writes `plots/limber_real`'s from that same derivative base plus one CosmoLike evaluation of the same model in the Limber approximation (run from `Cocoa/` with the cocoa environment). Each folder's `plot_<name>.py` redraws its figure from that file alone.

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